



REPRODUCTION AND ENDOCRINE SYSTEM

By Sanskriti

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Introduction to Reproduction

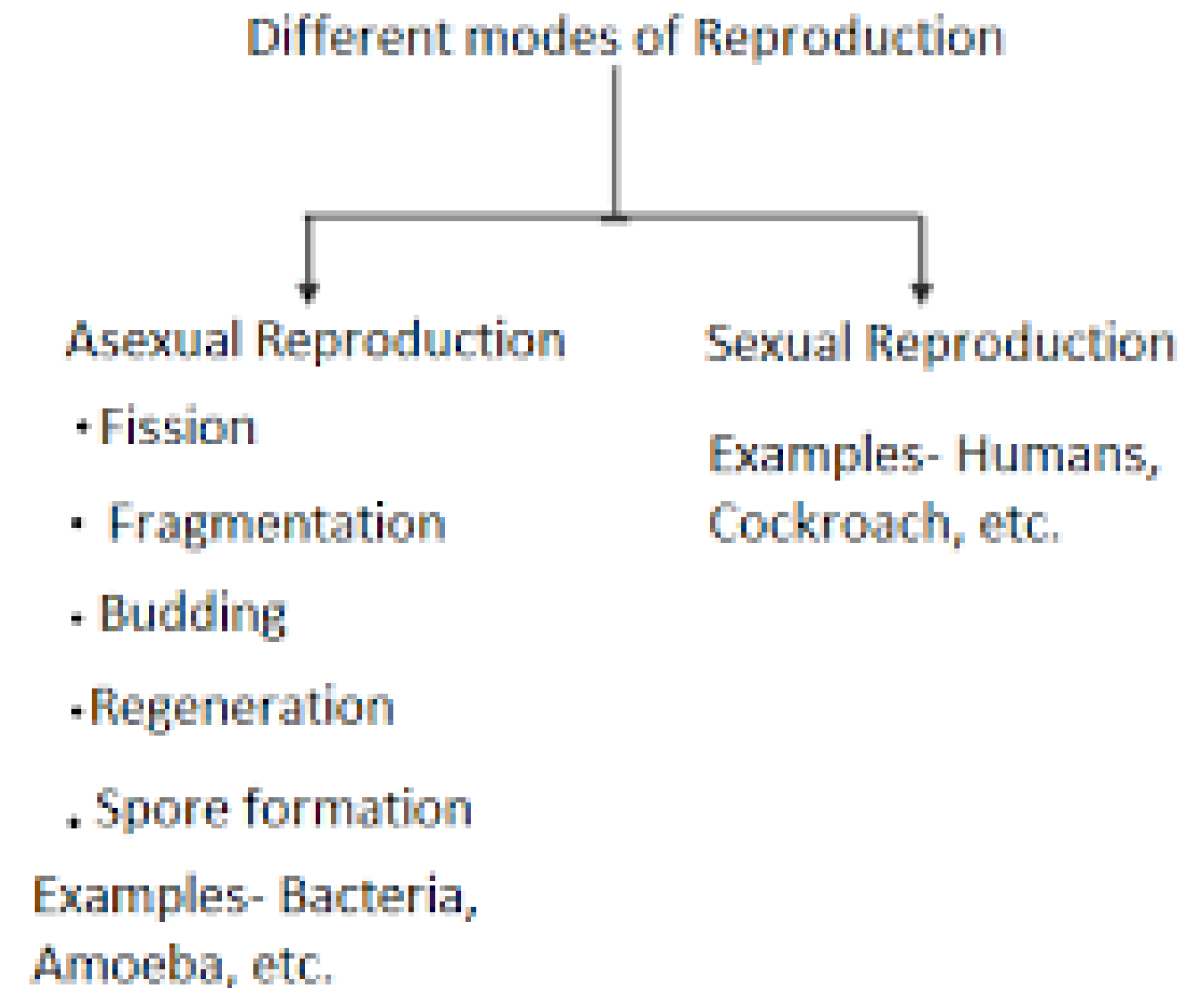
What is Reproduction?

- Reproduction is the process by which plants and animals produce young ones of their own kind
- It is one of the most important properties of living beings
- Ensures continuation of similar kinds of individuals, generation after generation

Types of Reproduction:

1. Asexual Reproduction - Single parent involved

2. Sexual Reproduction - Two parents involved (male and female)



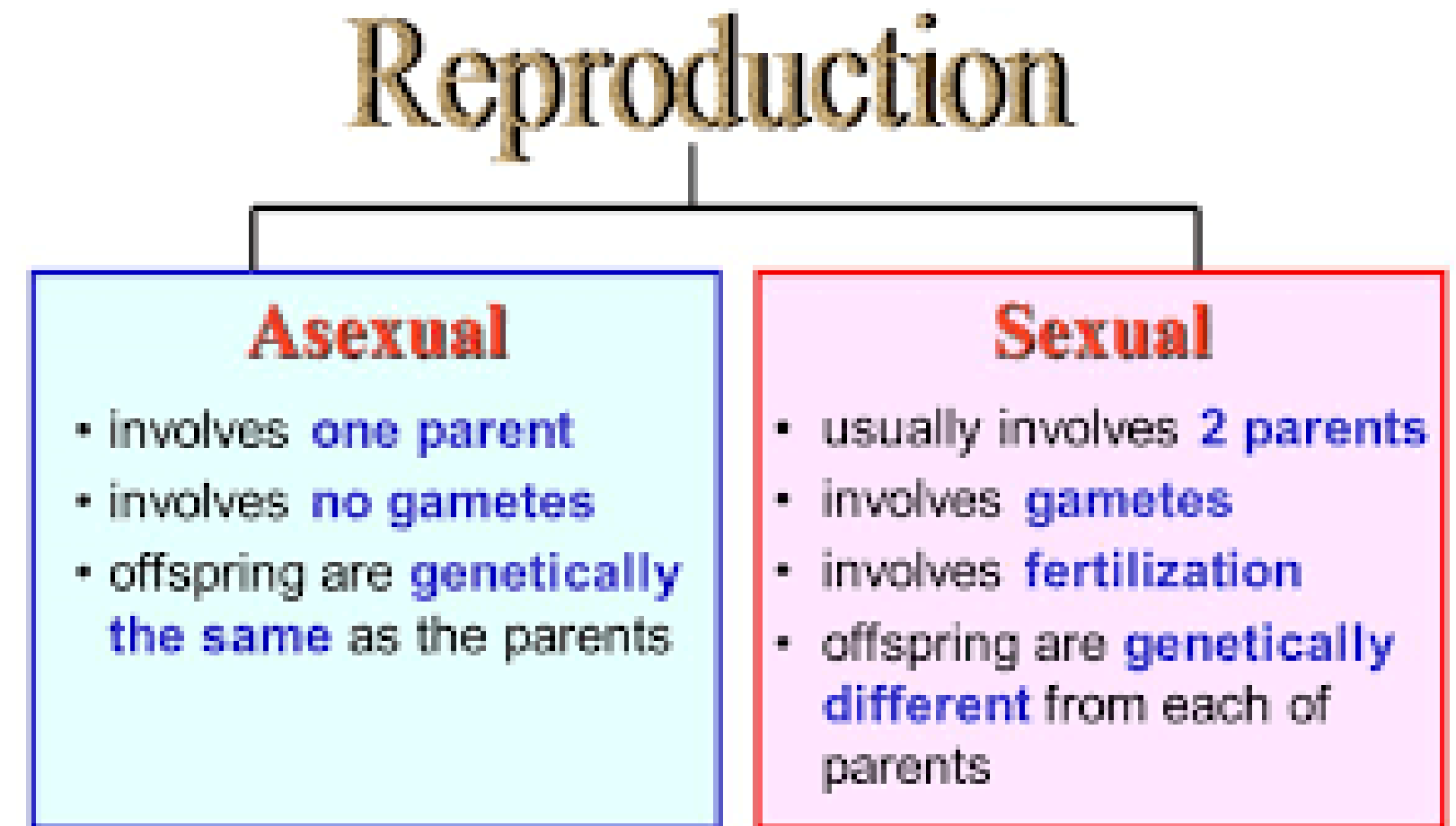
Asexual vs Sexual Reproduction

Asexual Reproduction:

- Only one parent required
- No fusion of male and female gametes
- Offspring are genetically identical to parent
- Examples: Budding in Hydra, Binary fission in Amoeba

Sexual Reproduction:

- Two parents required (male and female)
- Fusion of male and female gametes occurs
- Offspring show variation from parents
- Most common in higher animals



Introduction to Reproduction

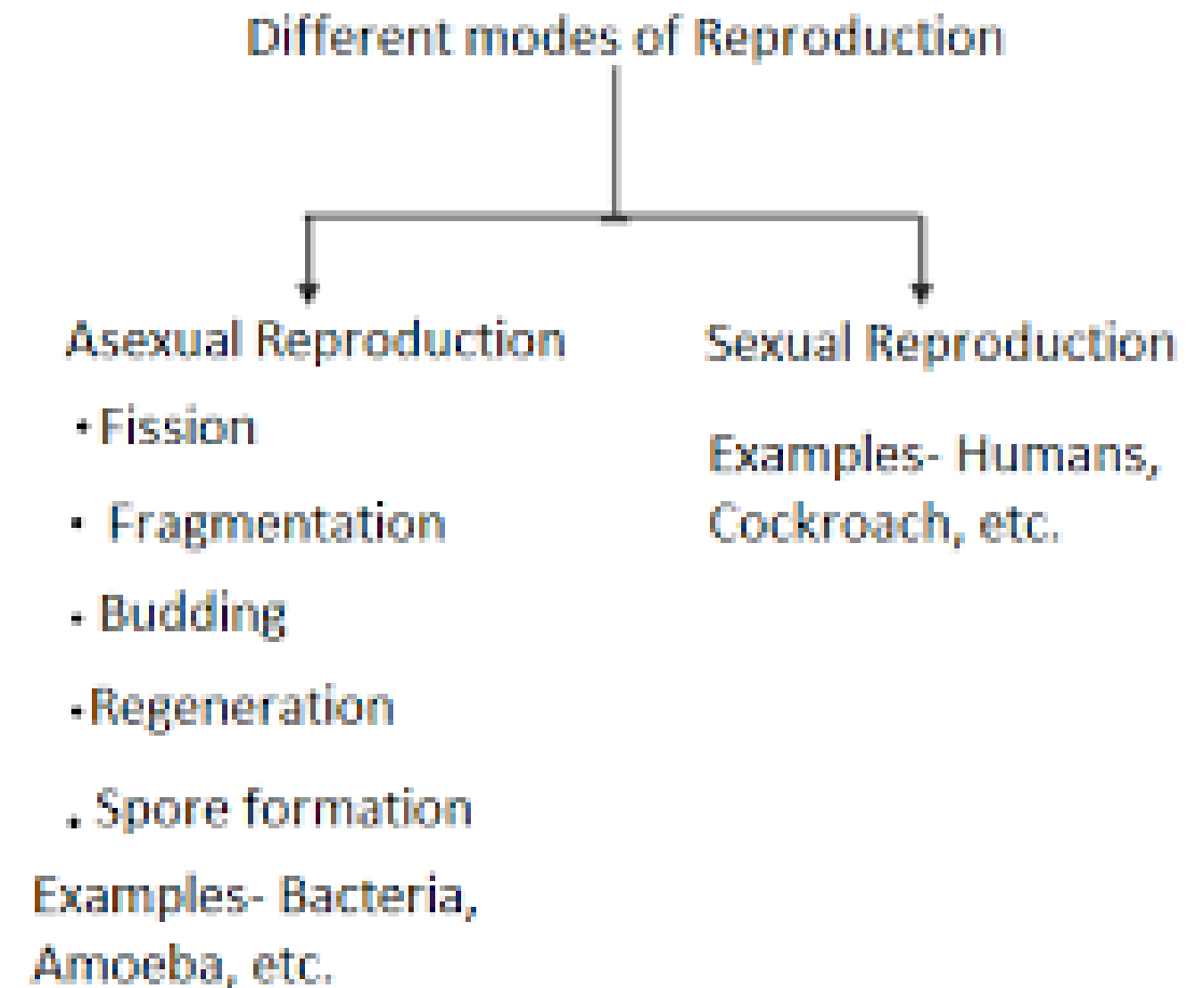
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Types of Reproduction:

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2. Sexual Reproduction - Two parents involved (male and female)



Introduction to Asexual Reproduction

Definition:

Asexual Reproduction is the process by which organisms produce offspring from a single parent without the involvement of gametes (sex cells) or fertilization.

Key Characteristics:

- **Single parent involved** - No need for male and female
- **No gamete formation** - No sperm or egg cells required
- **No fertilization** - Direct reproduction from parent
- **Genetically identical offspring** - Clones of the parent
- **Rapid reproduction** - Quick multiplication of organisms
- **Energy efficient** - Less energy required compared to sexual reproduction

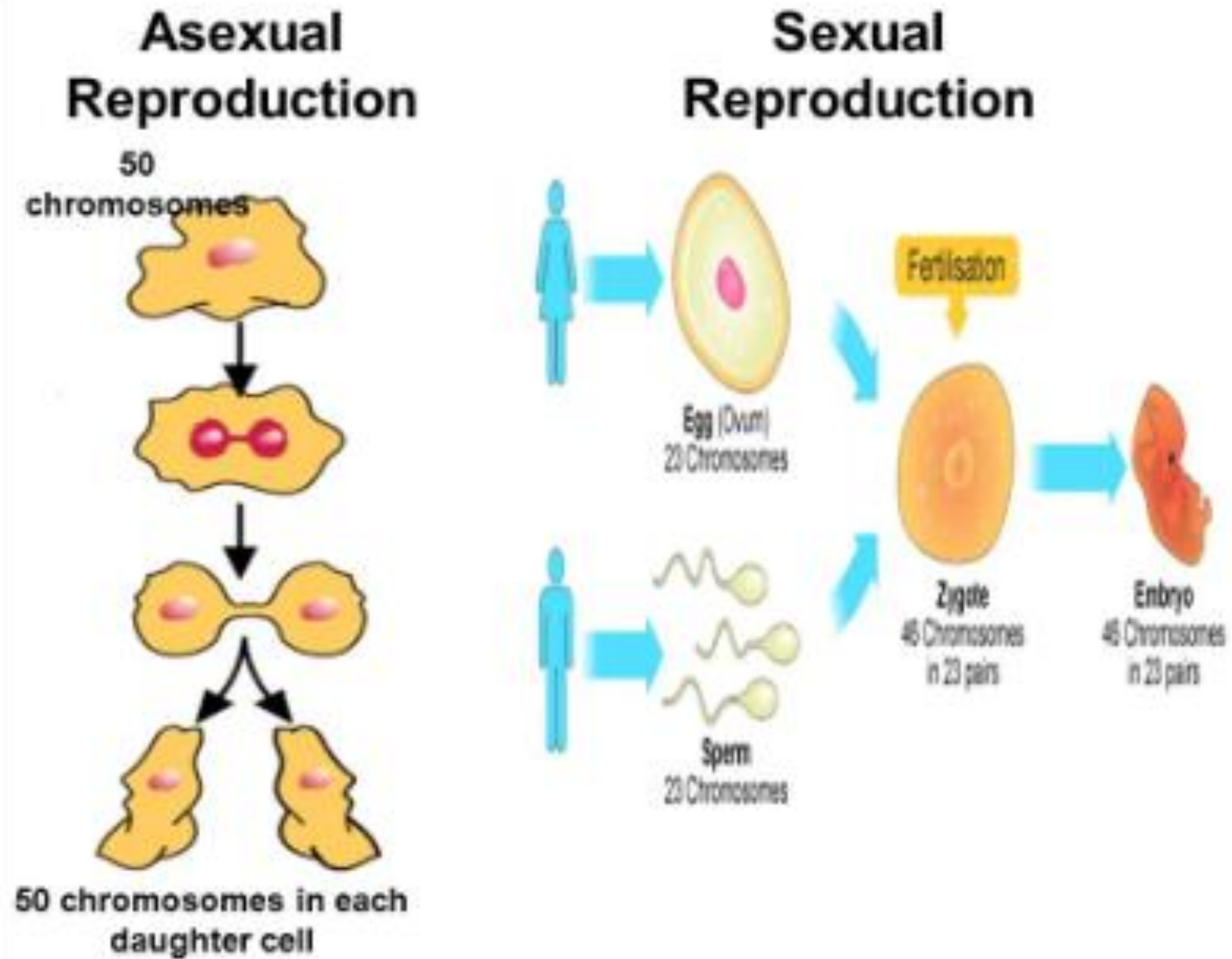
Advantages:

- Fast population growth
- No need to find a mate
- Favorable traits preserved exactly
- Efficient in stable environments
- Less energy expenditure

Disadvantages:

- No genetic variation
- Vulnerable to environmental changes
- Disease susceptibility increases
- No evolution through genetic recombination

Introduction to Reproduction



FISSION

Definition:

Fission is the splitting of a parent organism into two or more offspring of approximately equal size.

Types of Fission:

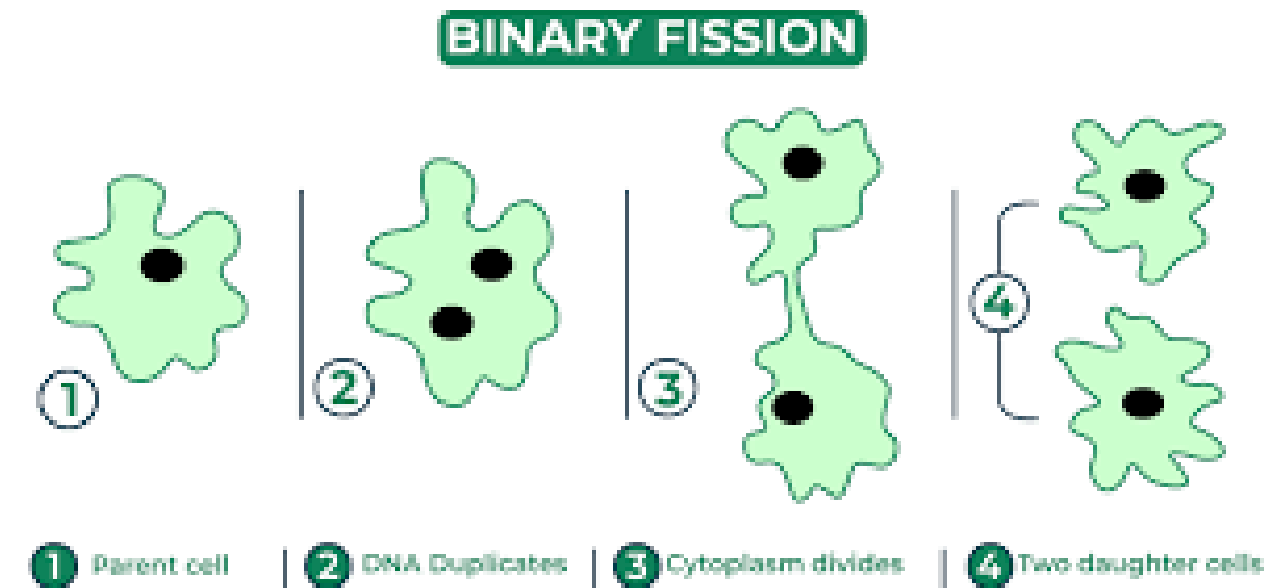
A) Binary Fission (Split into 2)

Process:

1. DNA replication occurs
2. Cell grows in size
3. Nuclear material divides
4. Cytoplasm divides
5. Two identical organisms formed

Examples:

- **Amoeba** - Irregular division plane
- **Paramecium** - Transverse division (across width)
- **Euglena** - Longitudinal division (along length)
- **Bacteria** - Simple binary fission



FISSION

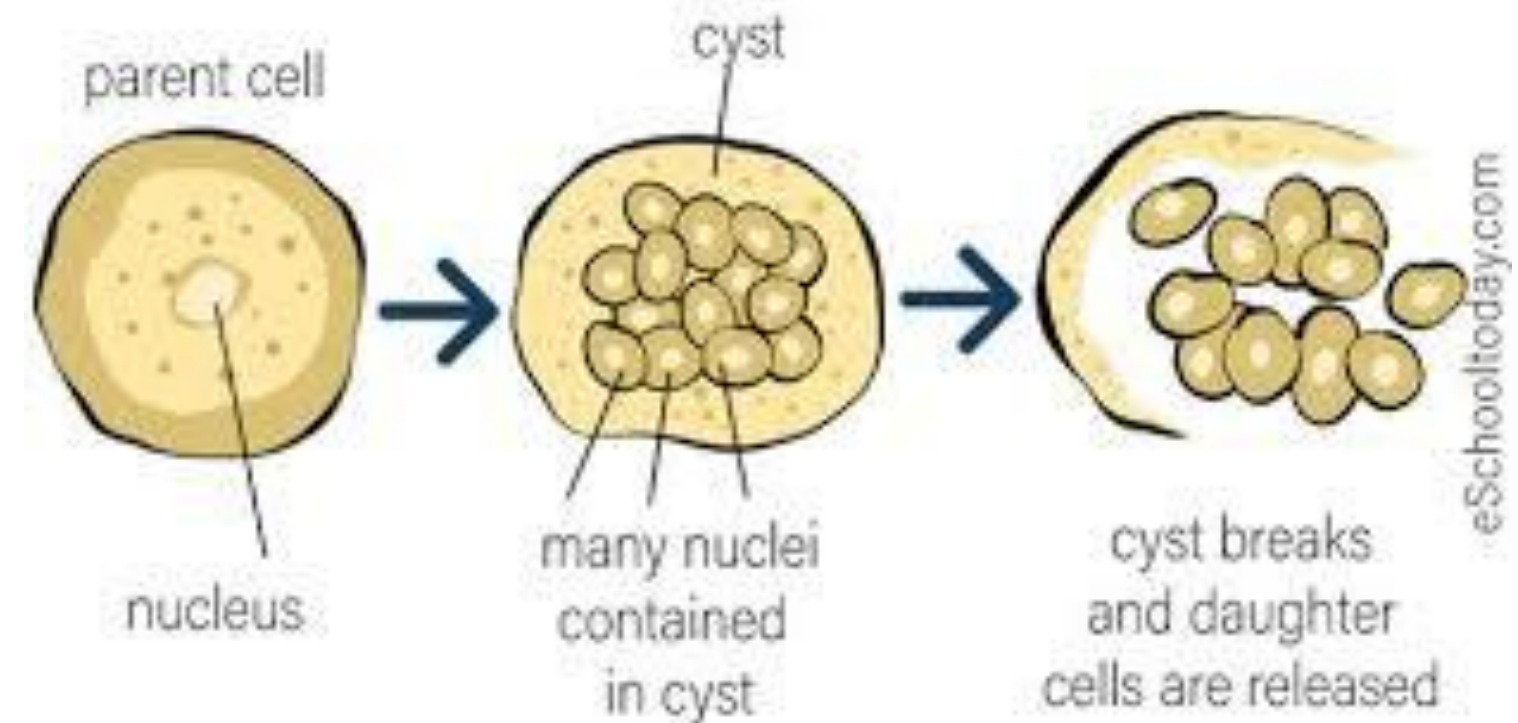
B) Multiple Fission (Split into many)

Process:

1. Organism forms protective cyst
2. Nucleus divides multiple times
3. Cytoplasm divides around each nucleus
4. Many small organisms formed
5. Released when conditions are favorable

Examples:

- **Plasmodium** (malaria parasite)
- **Amoeba** (during unfavorable conditions)



BUDDING

BUDDING

Definition:

Budding is the formation of a new individual from a small outgrowth (bud) that develops on the parent body.

Process:

1. Bud initiation - Small outgrowth appears on parent

2. Bud growth - Outgrowth increases in size

3. Organ development - Bud develops all necessary structures

4. Maturation - Bud becomes capable of independent life

5. Detachment - Bud may separate or remain attached

Examples:

Animals:

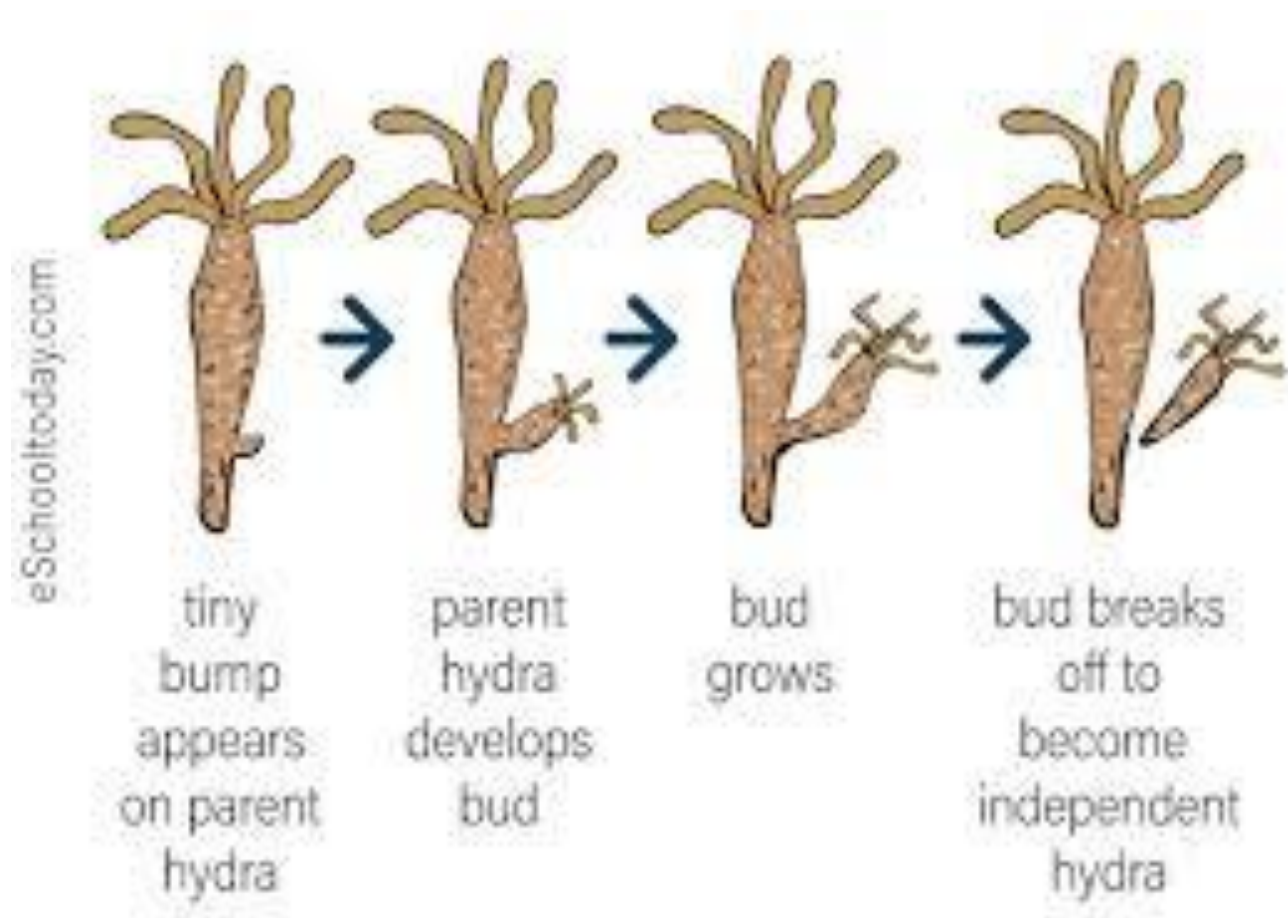
- **Hydra** - Small freshwater animal
 - Buds form on sides of body
 - Develop tentacles and mouth
 - Eventually detach from parent
- **Sponges** - Marine animals
 - Multiple buds form simultaneously
 - Create colonies when buds remain attached
- **Corals** - Marine cnidarians
 - Budding creates coral reefs
 - Thousands of individuals in one colony

Microorganisms:

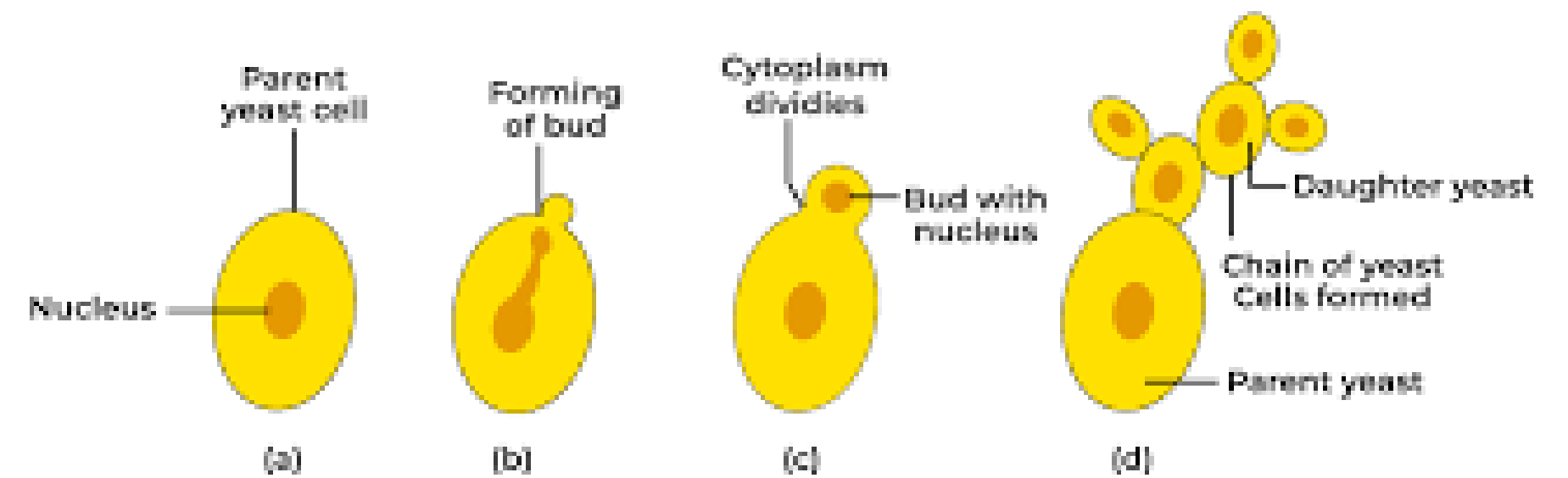
- **Yeast** - Unicellular fungus
 - Small bud forms on cell surface
 - Nucleus divides into bud
 - Bud separates when mature
 - Can form chains of connected cells

Advantages:

- Rapid reproduction
- Colony formation possible
- Parent continues to live and reproduce



Budding in Yeast



Regeneration

Definition:

Regeneration is the ability to regrow lost or damaged body parts, which can sometimes result in the formation of complete new organisms.

Process:

1. **Injury/Fragmentation** - Body part is lost or cut
2. **Wound healing** - Cut surface closes
3. **Blastema formation** - Mass of undifferentiated cells forms
4. **Regeneration** - Missing parts regrow
5. **Complete organism** - Full function restored

Examples:

Complete Regeneration (New organisms formed):

•Planaria (flatworm)

- Cut into pieces
- Each piece regenerates missing parts
- One worm becomes many worms

•Starfish/Sea stars

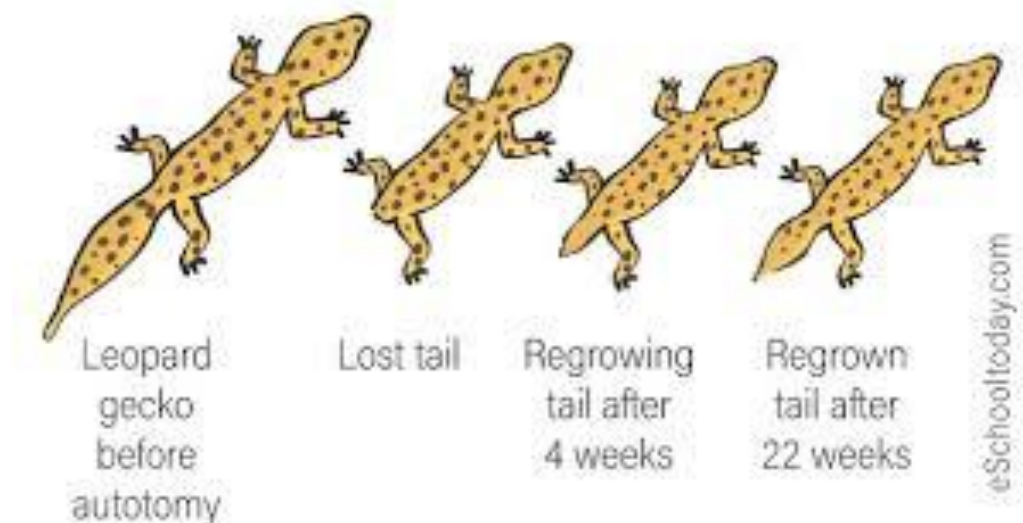
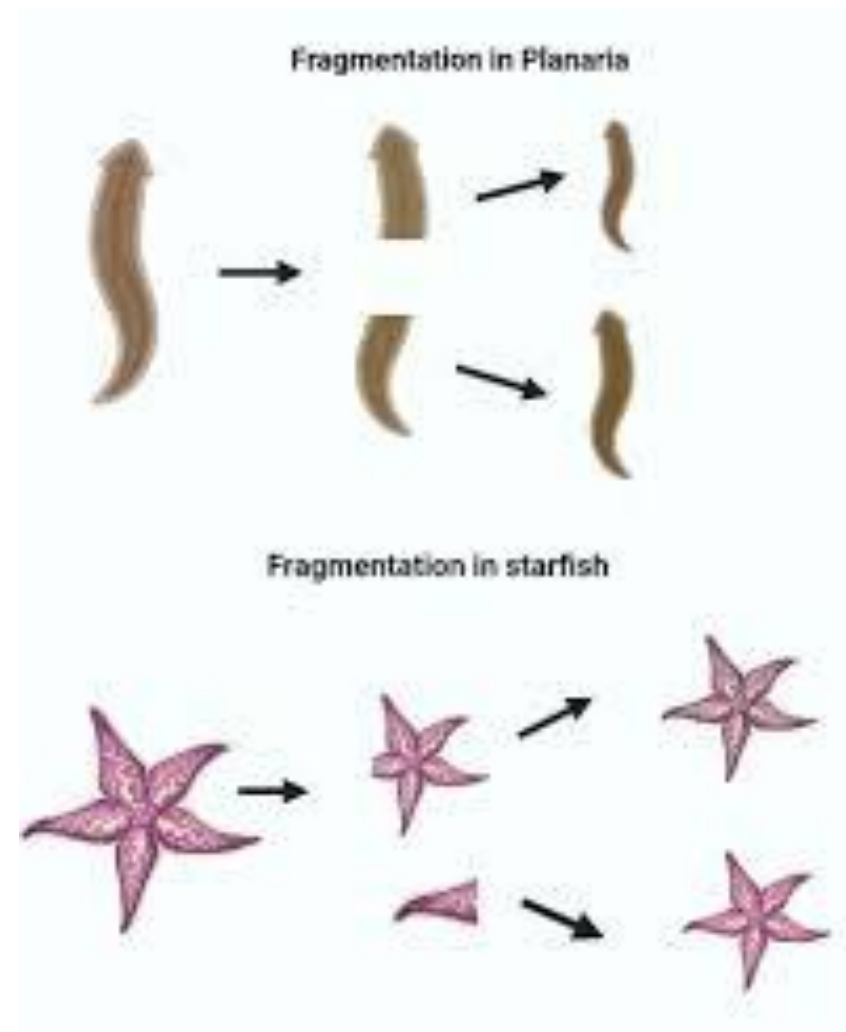
- Lost arms regrow
- Sometimes entire body from one arm
- Takes several months to complete

Partial Regeneration (Body parts only):

- **Lizards** - Tail regeneration
- **Crabs** - Claw regeneration
- **Salamanders** - Limb regeneration
- **Humans** - Liver regeneration (limited)

Biological Importance:

- Survival mechanism after injury
- Defense against predators
- Method of asexual reproduction in some species



FRAGMENTATION

Definition:

Fragmentation is the breaking of an organism into fragments, each capable of growing into a complete organism.

Difference from Regeneration:

- **Fragmentation:** Natural breaking for reproduction
- **Regeneration:** Usually occurs after accidental injury

Process:

1. **Natural breaking** - Organism breaks into pieces naturally
2. **Fragment dispersal** - Pieces spread to new locations
3. **Regeneration** - Each fragment grows missing parts
4. **New individuals** - Complete organisms develop

Examples:

Algae:

- **Spirogyra** - Green filamentous algae
 - Long chains break into smaller fragments
 - Each fragment grows into new filament
 - Common in ponds and streams
- **Sea lettuce** - Marine green algae
 - Thallus fragments break off
 - Each piece grows into new plant

Animals:

- **Planaria** - Can fragment naturally
- **Some annelid worms** - Break into segments
- **Coral polyps** - Fragments form new colonies

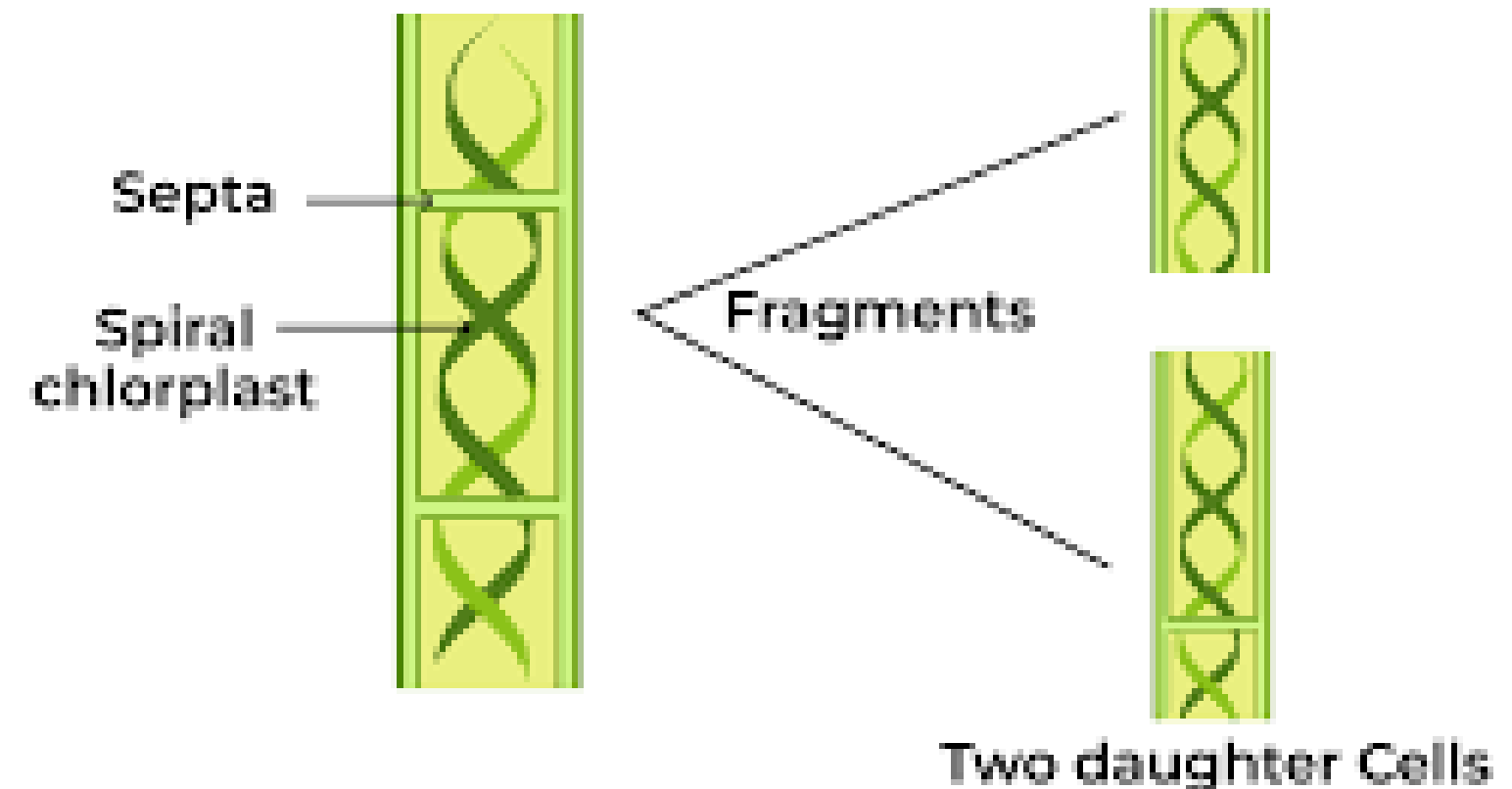
Fungi:

- **Rhizopus** - Hyphae break into fragments
- **Penicillium** - Mycelium fragmentation

Environmental Advantage:

- Dispersal to new habitats
- Colonization of suitable areas
- Population spread

Fragmentation in spirogyra



Vegetative propagation

Definition:

Vegetative propagation is reproduction through vegetative parts (roots, stems, leaves) of plants without involving seeds or flowers.

Types Based on Plant Parts:

A) Root-Based Propagation:

Examples:

- **Sweet Potato** - Adventitious buds on storage roots
- **Dahlia** - Tuberous roots with buds
- **Carrot** - Can regenerate from root pieces
- **Radish** - Root cuttings

B) Stem-Based Propagation:

Underground Stems:

- **Rhizomes** - Horizontal underground stems
 - Ginger, Turmeric, Bamboo, Iris
 - Nodes produce new shoots and roots
- **Tubers** - Swollen underground stems
 - Potato - "Eyes" are actually buds
 - Each eye can grow into new plant
- **Bulbs** - Short stems with fleshy leaves
 - Onion, Garlic, Tulip, Daffodil
 - Bulblets form around main bulb
- **Corms** - Swollen underground stems
 - Gladiolus, Crocus, Taro

Above-ground Stems:

- **Runners/Stolons** - Horizontal stems above ground
 - Strawberry, Grass, Mint
 - New plants form at nodes
- **Suckers** - Shoots from roots or stem base
 - Banana, Rose, Bamboo

C) Leaf-Based Propagation:

- **Bryophyllum** - Plantlets form on leaf margins
- **Begonia** - Leaf cuttings root easily
- **Kalanchoe** - Adventitious buds on leaves
- **African violet** - Leaf petiole cuttings

Artificial Methods (Human-induced):

- **Cutting** - Rose, Sugarcane, Grapes
- **Grafting** - Fruit trees, Roses
- **Layering** - Jasmine, Rose
- **Tissue culture** - Orchids, Bananas

Advantages:

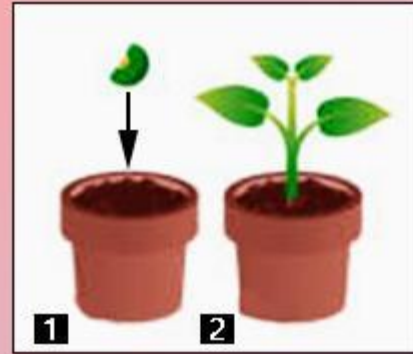
- Faster than seed germination
- Maintains exact parent characteristics
- Useful for seedless varieties
- Economically important in agriculture

Disadvantages:

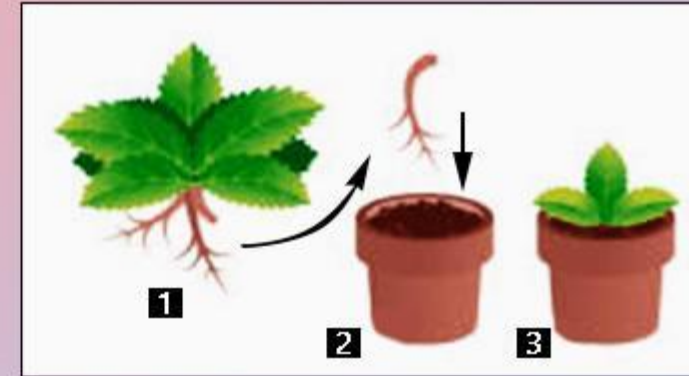
- No genetic variation
- Disease spreads easily
- May lead to overcrowding
- Soil depletion possible

TYPES OF VEGETATIVE PROPOGATION

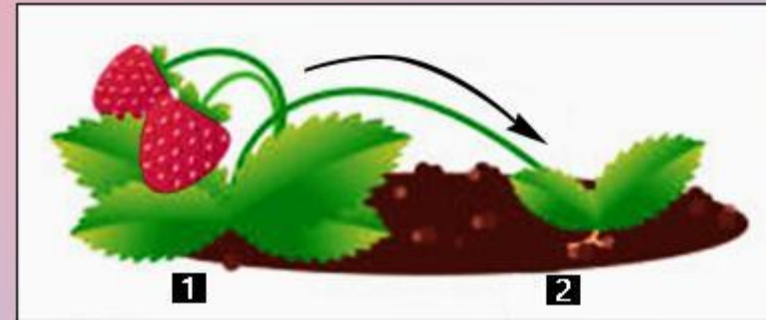
ASEXUAL SEEDS



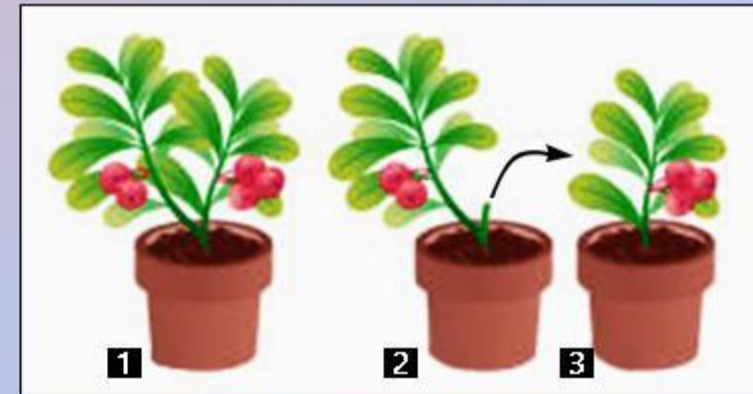
ROOT CUTTING



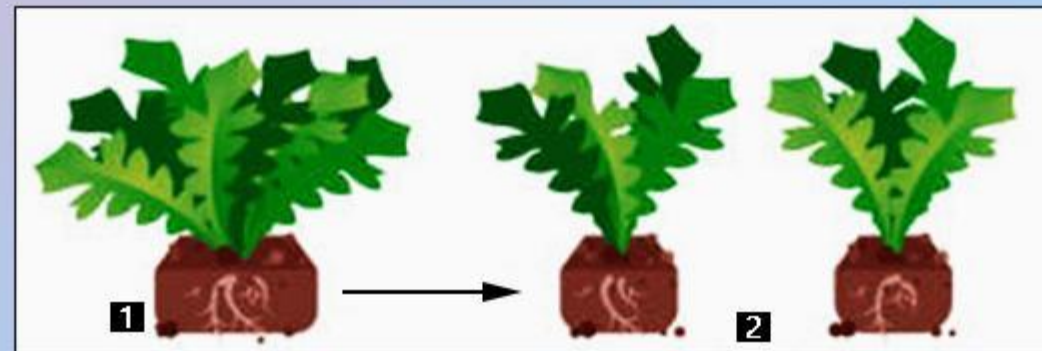
SUCKERS



GRAFTING



DIVISION



SPORE FORMATION

Examples:

Fungi:

- **Rhizopus** (Bread mold)
 - Black sporangia on stalks
 - Thousands of spores per sporangium
 - Spores germinate into new mycelium
- **Penicillium** - Blue-green spores (conidia)
- **Mushrooms** - Spores from gills or pores
- **Yeast** - Ascospores in tough-walled asci

Plants:

- **Ferns** - Spores in sori (clusters) under leaves
- **Mosses** - Spores in capsules on stalks
- **Algae** - Various types of spores

Bacteria:

- **Bacillus** - Endospores for survival
- **Clostridium** - Heat-resistant spores

Types of Spores:

1. **Zoospores** - Motile spores with flagella (Chlamydomonas)
2. **Aplanospores** - Non-motile spores (Chlorella)
3. **Endospores** - Internal bacterial spores
4. **Conidia** - External fungal spores
5. **Ascospores** - Spores in sac-like structures
6. **Basidiospores** - Spores from mushroom-type fungi

Dispersal Mechanisms:

- **Wind dispersal** - Light, small spores
- **Water dispersal** - Motile spores
- **Animal dispersal** - Sticky or hooked spores
- **Explosive release** - Sporangia burst forcefully

Survival Strategy:

- Survive extreme temperatures
- Resist desiccation (drying)
- Remain viable for years
- Germinate when conditions improve

Definition:

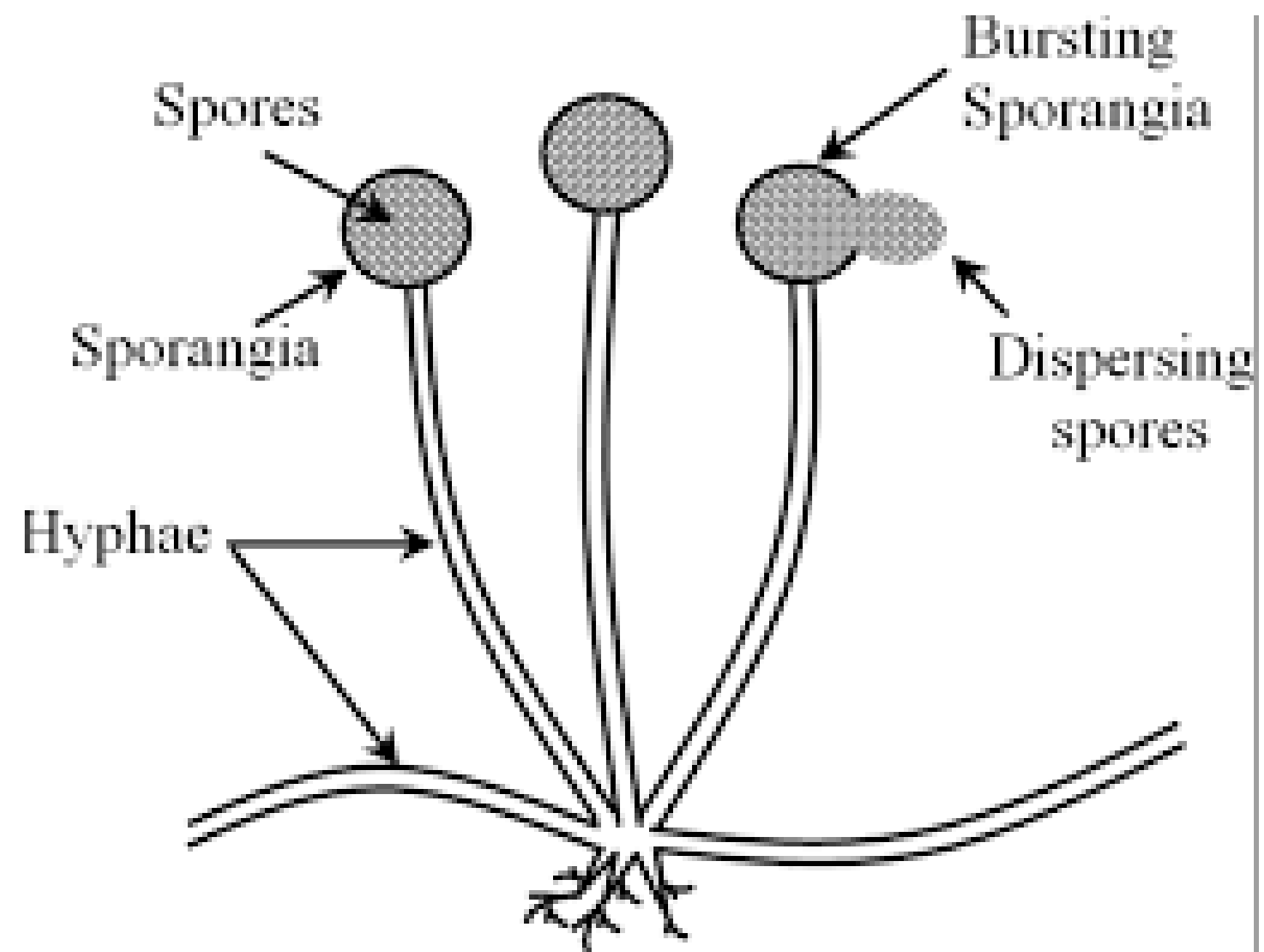
Spore formation is the production of specialized reproductive cells called spores that can develop into new organisms under favorable conditions.

Characteristics of Spores:

- **Resistant structure** - Withstand harsh conditions
- **Lightweight** - Easy wind/water dispersal
- **Protective wall** - Prevents water loss
- **Dormant** - Remain inactive until conditions improve
- **Single-celled** - Simple reproductive unit

Process:

1. **Spore development** - Formed in specialized structures (sporangia)
2. **Maturation** - Spores develop protective coating
3. **Release** - Sporangia burst, releasing spores
4. **Dispersal** - Wind, water, or animals carry spores
5. **Germination** - Under favorable conditions, spores grow
6. **Development** - New organism develops from spore



Type	Speed	Organisms	Environment	Offspring Number
Fission	Very Fast	Unicellular	Favorable	2 or many
Budding	Fast	Simple multicellular	Stable aquatic	Multiple
Regeneration	Slow-Medium	Complex animals	After injury	Few
Fragmentation	Medium	Algae, simple animals	Natural breaking	Multiple
Vegetative Propagation	Medium	Plants	Seasonal	Multiple
Spore Formation	Variable	Fungi, plants, bacteria	Harsh conditions	Thousands

Introduction to Plant Reproduction

Definition:

Plant reproduction is the process by which plants produce new individuals of their own kind to continue their species.

Types of Reproduction in Plants:

1.Asexual Reproduction - Single parent involved, no gametes

2.Sexual Reproduction - Two parents involved, gametes fuse

Asexual Reproduction

One parent

No gamete formation

Offspring identical to parent

Faster process

Vegetative parts involved

Sexual Reproduction

Two parents (male & female gametes)

Gametes formed

Offspring show variation

Slower process

Reproductive organs (flowers) involved

Introduction to Reproduction

Importance:

- Species continuation
- Genetic diversity (in sexual reproduction)
- Adaptation to environment
- Economic importance in agriculture

Introduction to Reproduction

Definition:

Asexual reproduction in plants occurs through **vegetative propagation** - reproduction through vegetative parts like roots, stems, and leaves.

Characteristics:

- No flowers or seeds involved
- Single parent produces offspring
- Offspring are genetically identical (clones)
- Faster than sexual reproduction
- Common in favorable environments

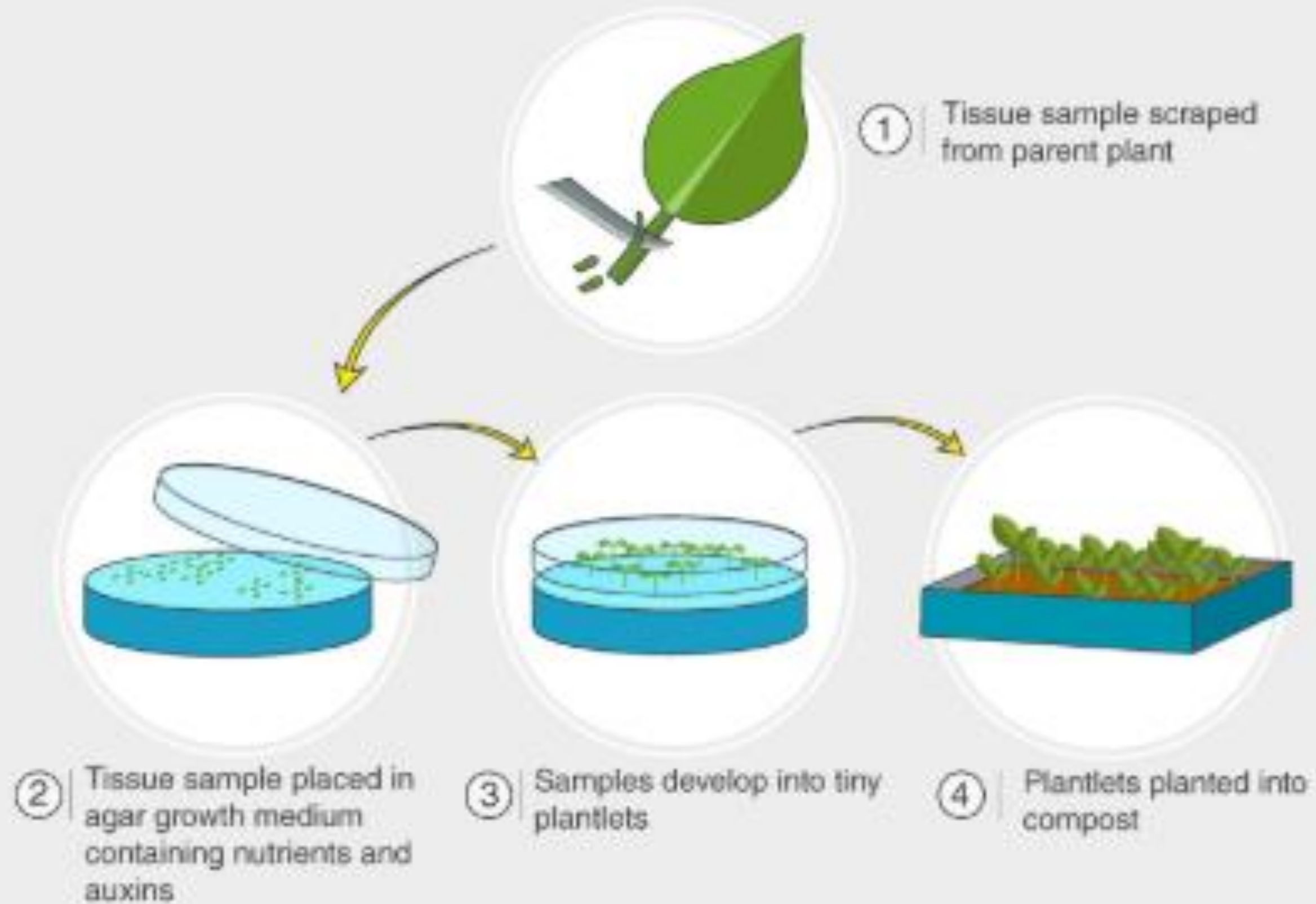
Natural Methods:

- 1.Root propagation
- 2.Stem propagation
- 3.Leaf propagation

Artificial Methods:

- 1.Cutting
- 2.Grafting
- 3.Layering
- 4.Tissue culture

PLANT TISSUE CULTURE



SIMPLE LAYERING



MOUND (STOOL) LAYERING



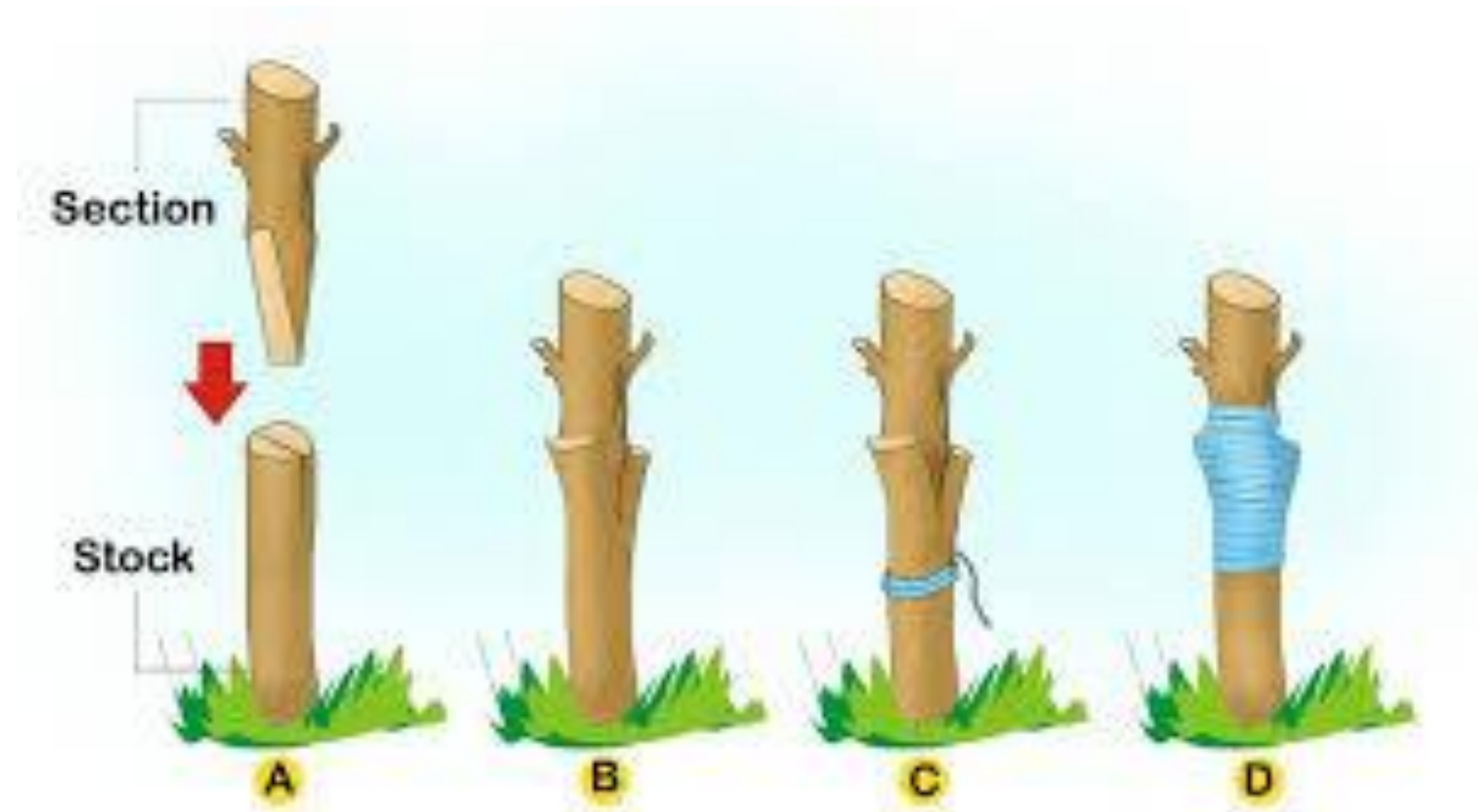




Fig 5.1 Cuttings need "spare" leaves at the base

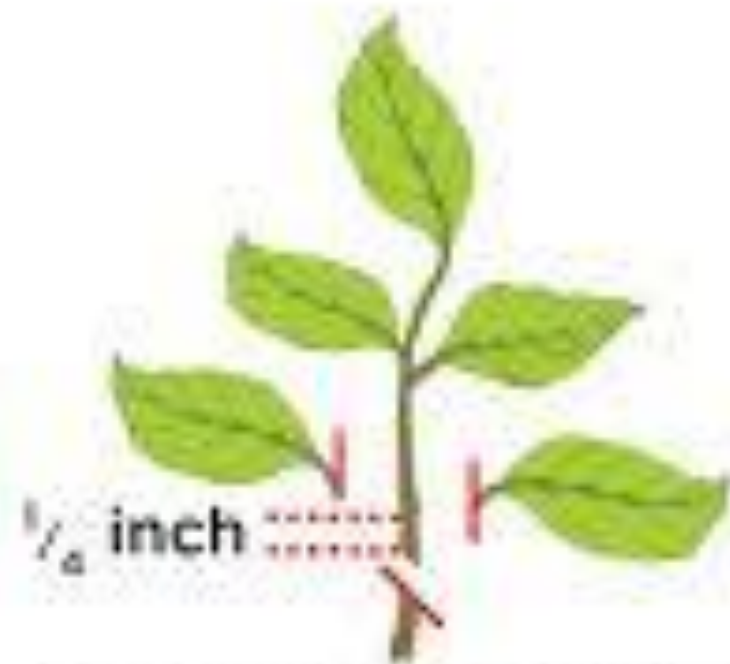


Fig 5.2 Where to cut



Fig 5.3 Dip cutting in FloraMax Cloner

1-inch deep



Fig 5.4 Insert base of cutting into substrate

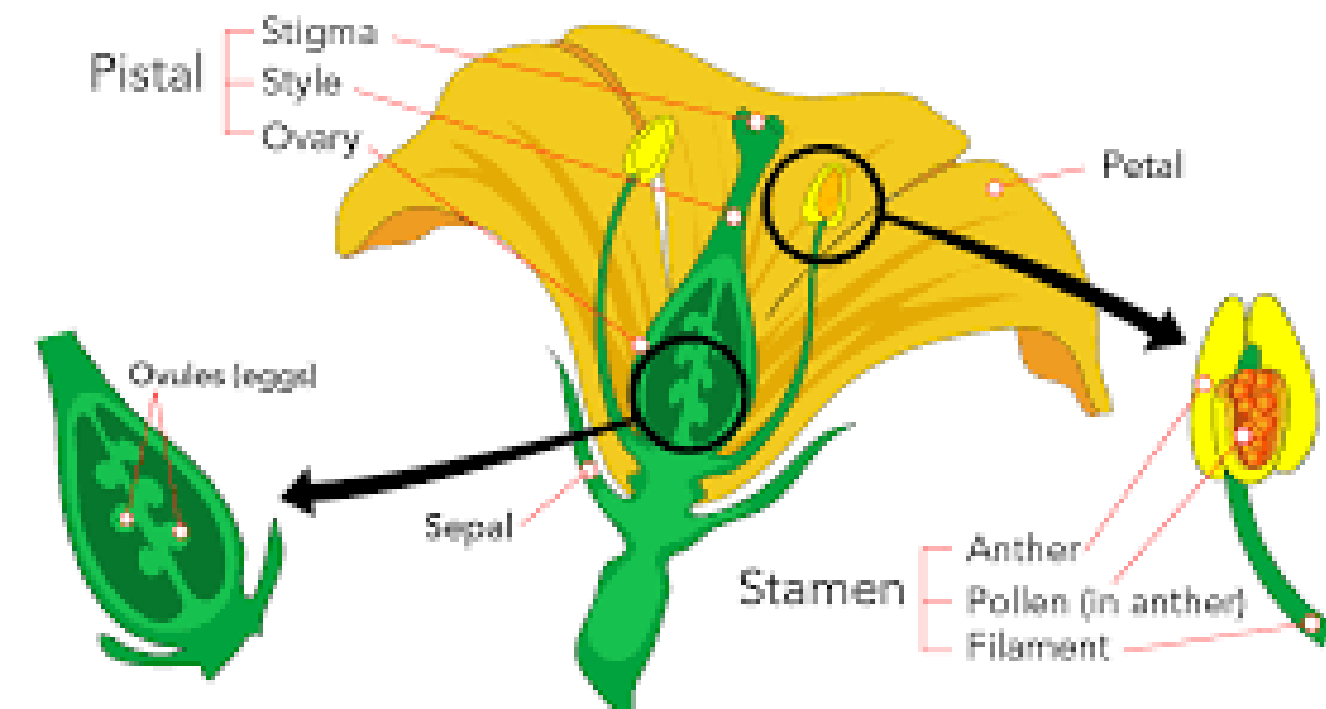
Sexual Reproduction in Plants

Definition:

Sexual reproduction in plants involves the **fusion of male and female gametes** to form offspring with genetic variation.

Key Features:

- **Flowers** are the reproductive organs
- **Pollination** transfers male gametes to female organs
- **Fertilization** occurs when gametes fuse
- **Seeds** are formed containing embryos
- **Fruits** develop to protect and disperse seeds



Sexual Reproduction in Plants

Male Parts (Androecium):

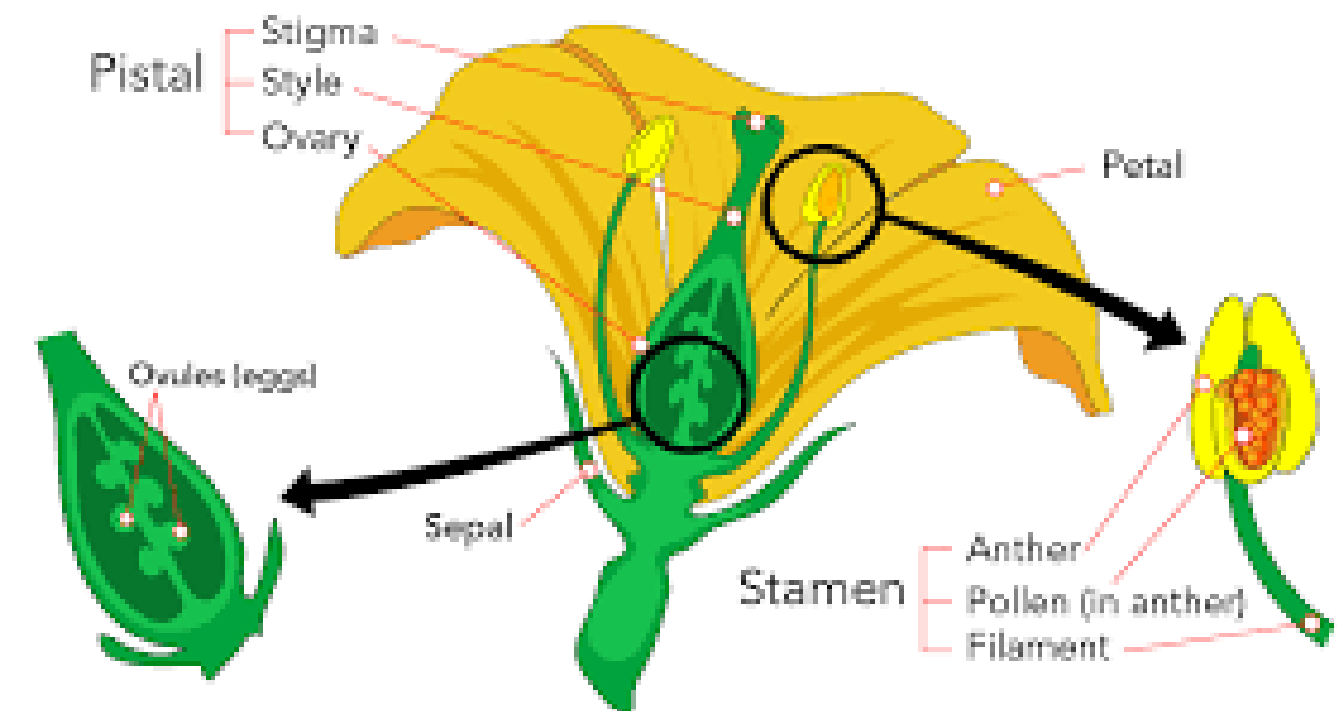
- **Stamen** - Male reproductive organ
- **Anther** - Produces pollen grains (male gametes)
- **Filament** - Stalk supporting anther

Female Parts (Gynoecium):

- **Pistil/Carpel** - Female reproductive organ
- **Stigma** - Receives pollen
- **Style** - Connects stigma to ovary
- **Ovary** - Contains ovules (female gametes)

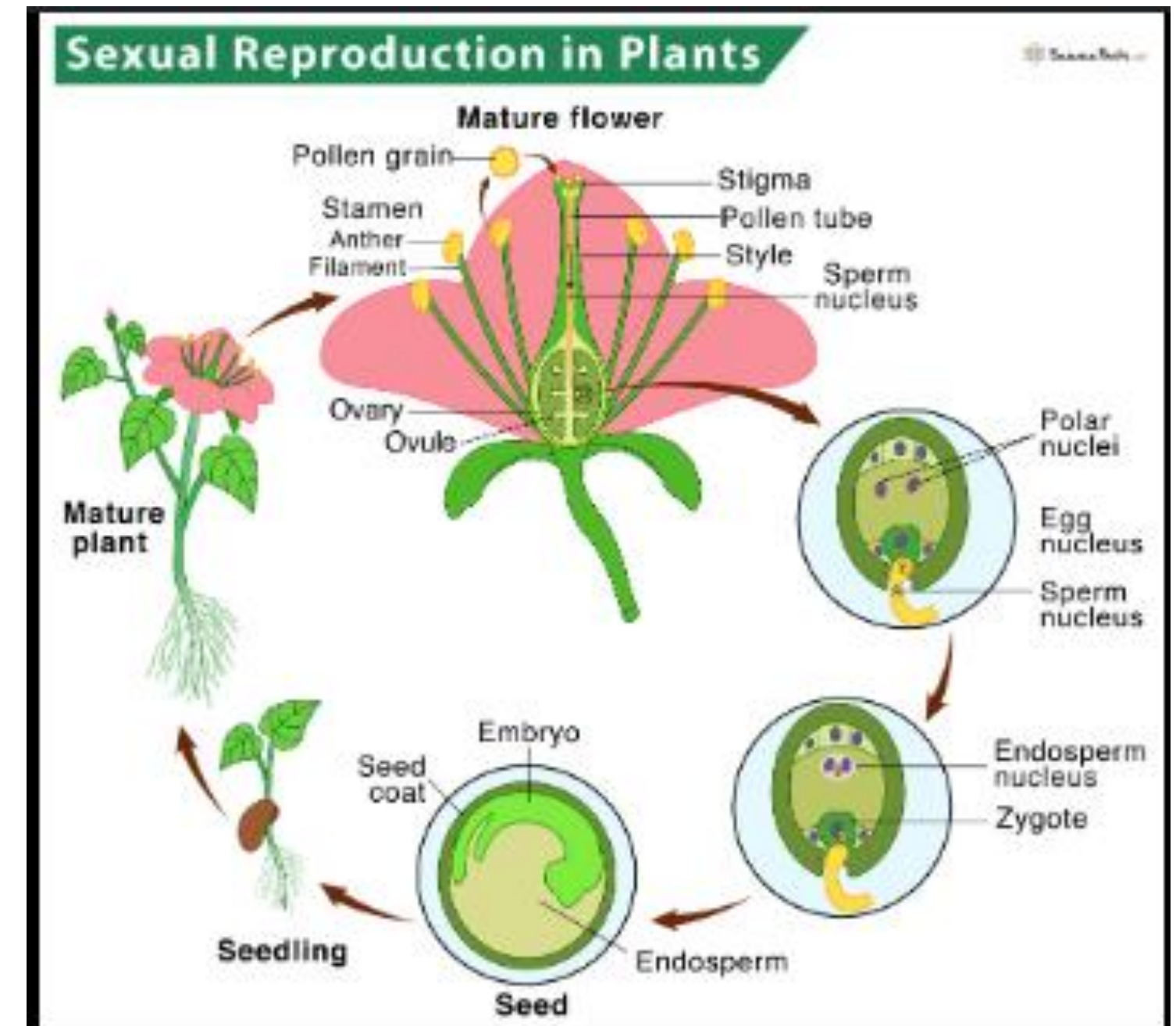
Accessory Parts:

- **Sepals** - Protect flower bud
- **Petals** - Attract pollinators



Introduction to Reproduction

- **Pollen formation** in anthers
- **Ovule formation** in ovary
- **Pollination** - pollen transfer
- **Fertilization** - gamete fusion
- **Seed formation**
- **Fruit development**



Introduction to Reproduction

Definition:

Pollination is the transfer of pollen grains from anther to stigma.

Types of Pollination:

1. Self-Pollination (Autogamy)

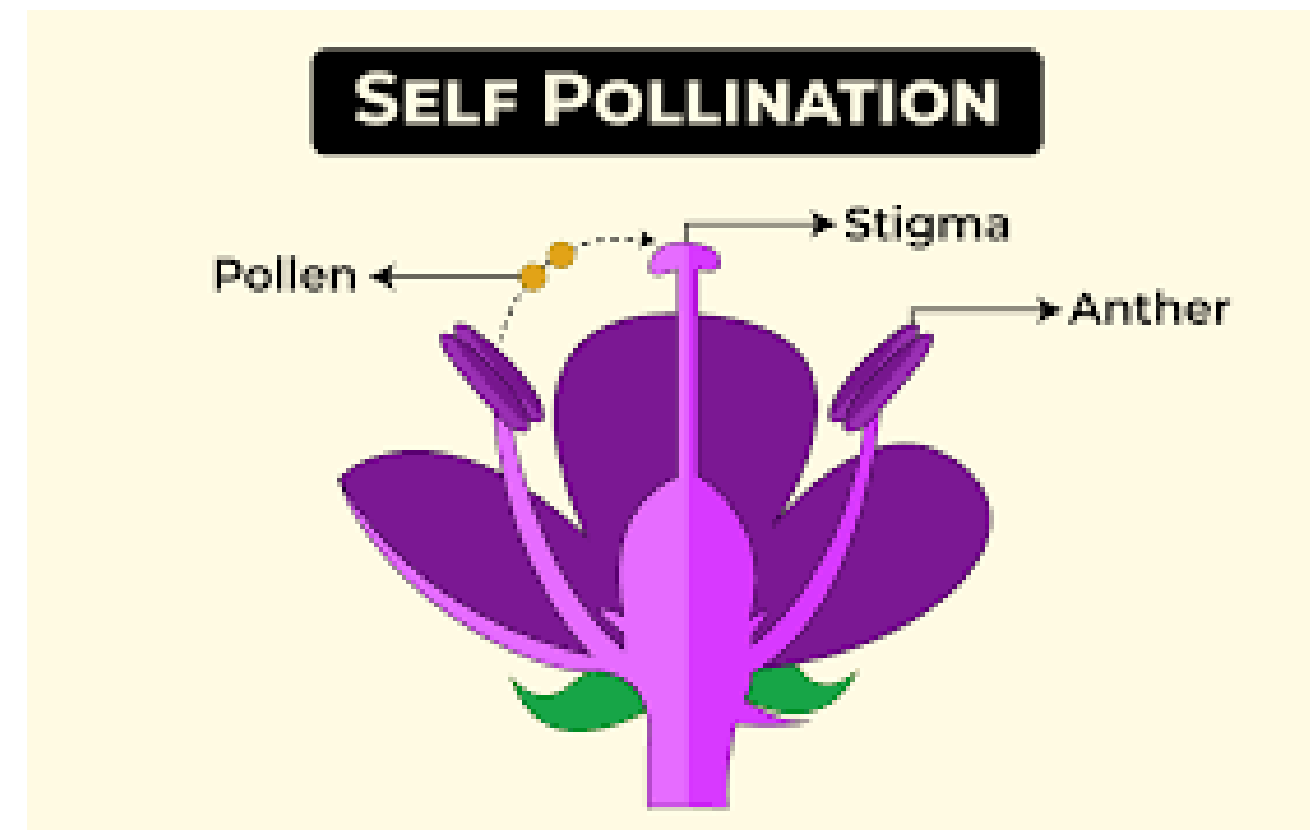
- Pollen from same flower reaches its own stigma
- Or pollen from one flower reaches stigma of another flower on same plant

Advantages:

- No dependence on external agents
- Maintains pure plant characteristics
- Possible even in isolation

Disadvantages:

- No genetic variation
- May lead to inbreeding depression
- Reduced adaptability



Introduction to Reproduction

2. Cross-Pollination (Allogamy)

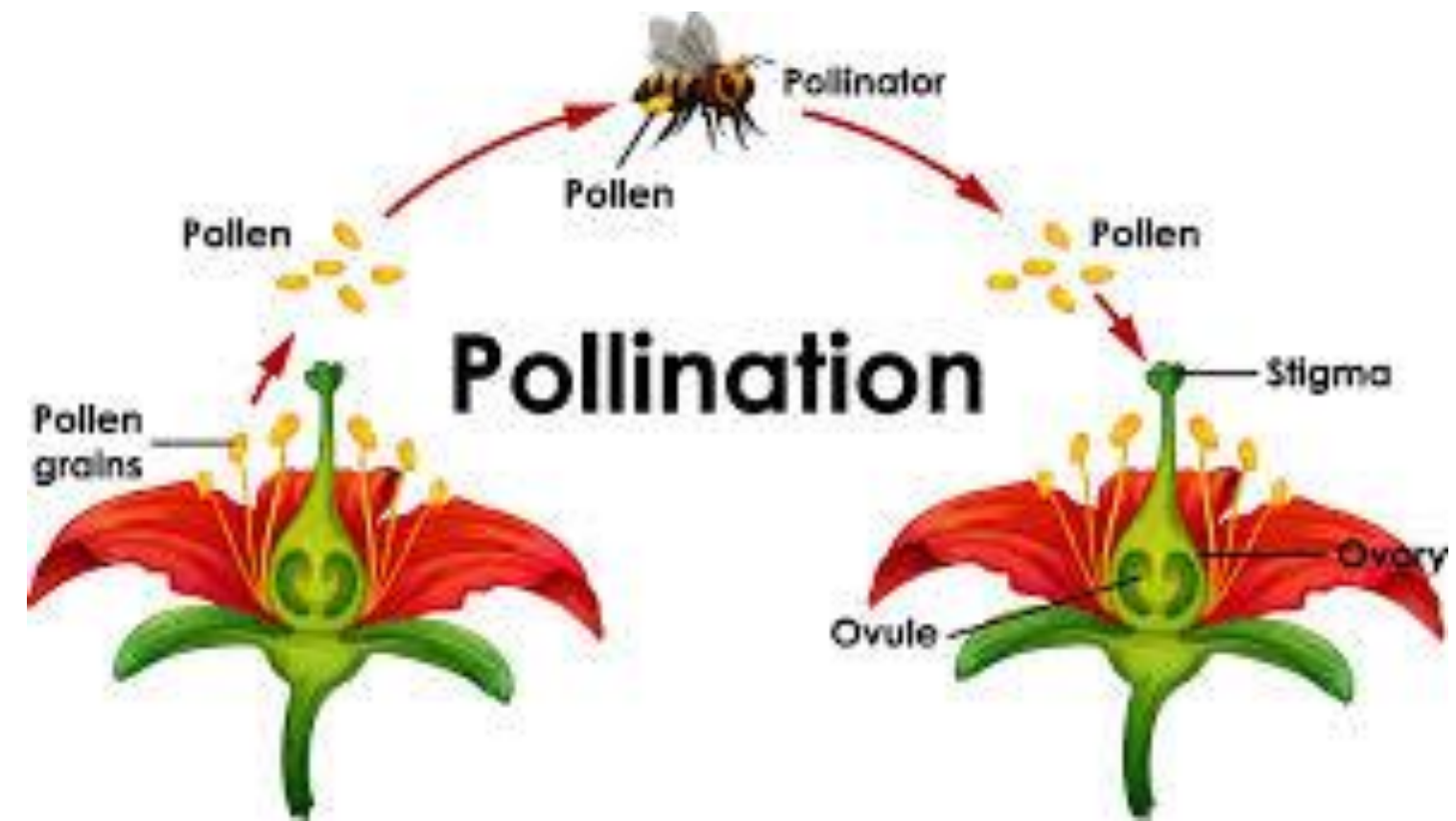
- Pollen from one plant reaches stigma of flower on different plant of same species

Advantages:

- Genetic variation in offspring
- Better adaptability
- Healthier offspring
- Evolution possible

Disadvantages:

- Depends on external agents
- Wastage of pollen
- No guarantee of pollination



Agents of Pollination:

Introduction to Reproduction

Abiotic Agents:

Wind (Anemophily):

- Examples: Grasses, corn, coconut
- Features: Small, light pollen; large stigma; no bright colors

Water (Hydrophily):

- Examples: Vallisneria, Hydrilla
- Features: Pollen carried by water currents

Biotic Agents:

Insects (Entomophily):

- Examples: Rose, sunflower, mustard
- Features: Bright colors, fragrance, nectar

Birds (Ornithophily):

- Examples: Bottle brush, coral tree
- Features: Red/orange colors, large flowers

Bats (Chiropterophily):

- Examples: Some cacti, baobab
- Features: Night blooming, strong fragrance

Modes of Reproduction in Animals

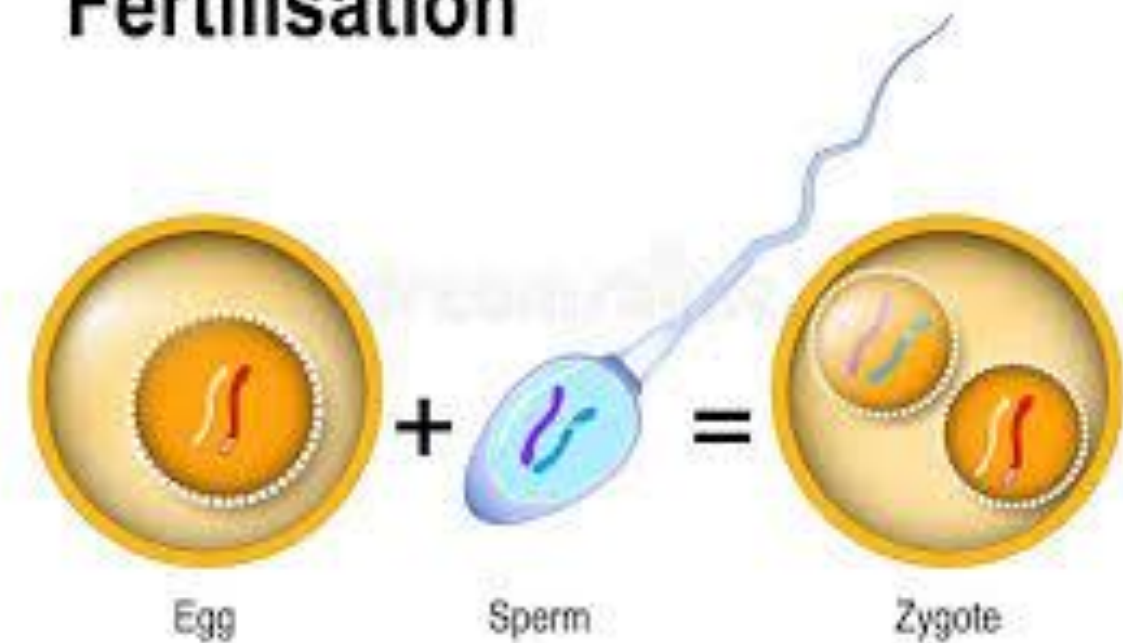
Sexual Reproduction involves:

- **Fertilization** - Union of male and female gametes
- **Development** - Growth of fertilized egg into new individual

Key Terms:

- **Gametes** - Reproductive cells (sperm and ovum)
- **Zygote** - Fertilized egg formed after fusion of gametes
- **Embryo** - Early stage of development after fertilization

Fertilisation



Sex Determination and Chromosomes

How is sex determined?

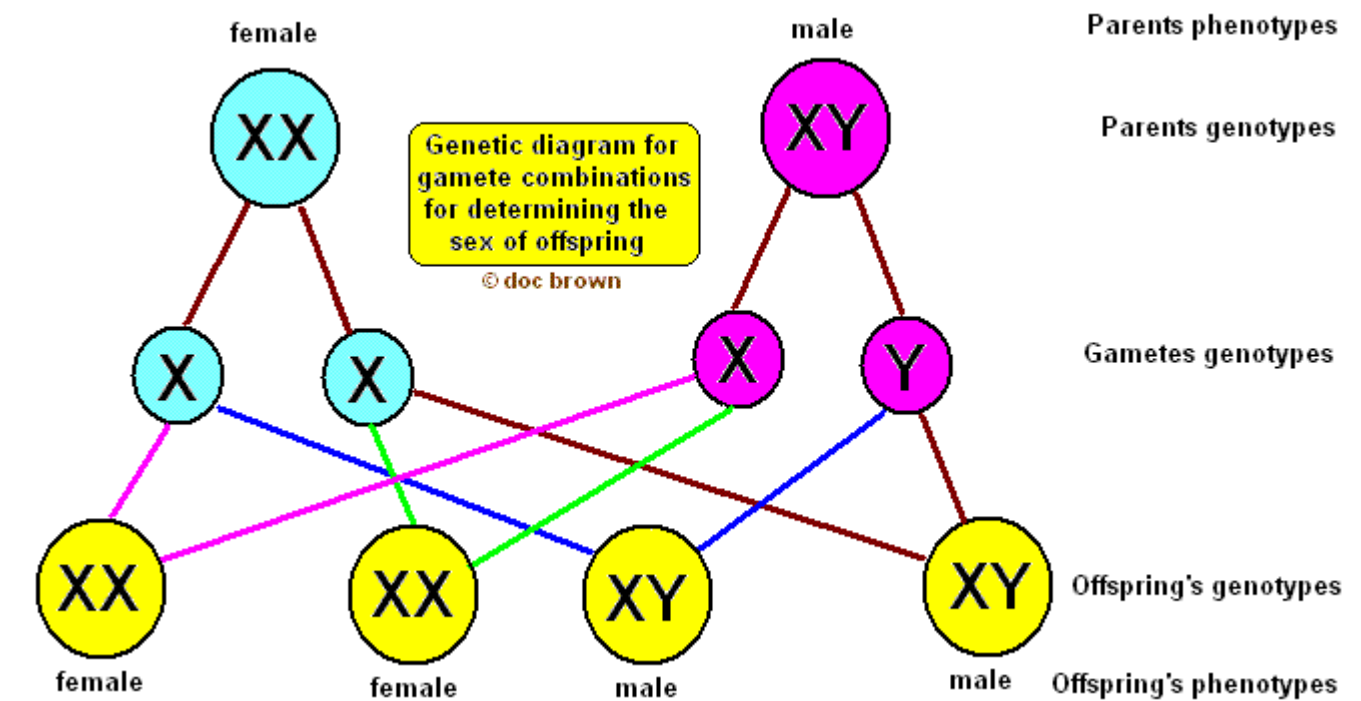
- Sex is determined by special chromosomes called sex chromosomes
- Humans have 23 pairs of chromosomes (46 total)
- 22 pairs are autosomes, 1 pair are sex chromosomes

Sex Chromosomes:

- **Females:** XX chromosomes
- **Males:** XY chromosomes
- During reproduction, each parent contributes one sex chromosome

Inheritance Pattern:

- Mother always gives X chromosome
- Father gives either X (girl child) or Y (boy child)
- Therefore, father determines the sex of the child



Gamete Formation

Male Gametes (Sperm):

- Produced in testes
- Small, motile cells
- Contain either X or Y chromosome
- Millions produced daily

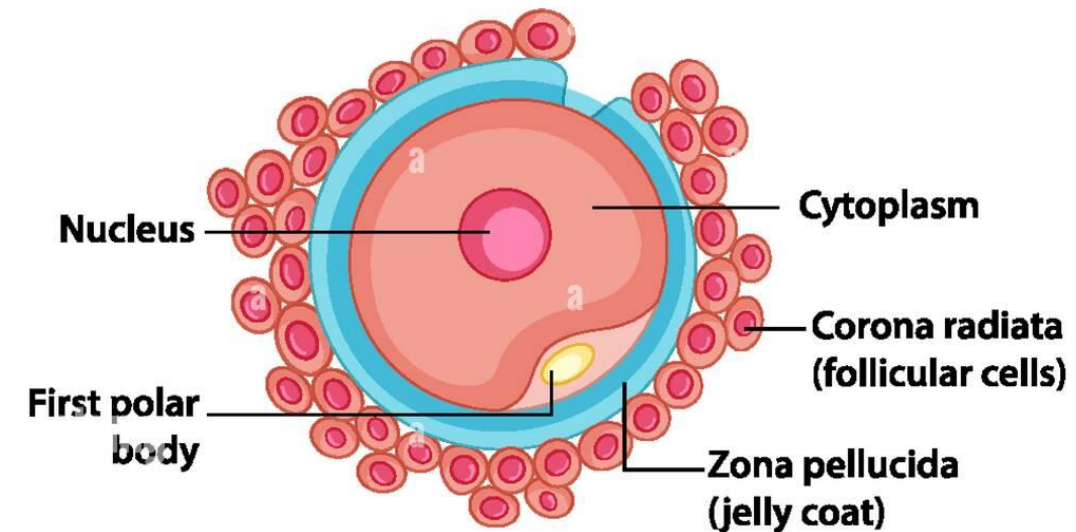
Female Gametes (Ovum/Egg):

- Produced in ovaries
- Large, non-motile cells
- Always contain X chromosome
- One released per month (in humans)

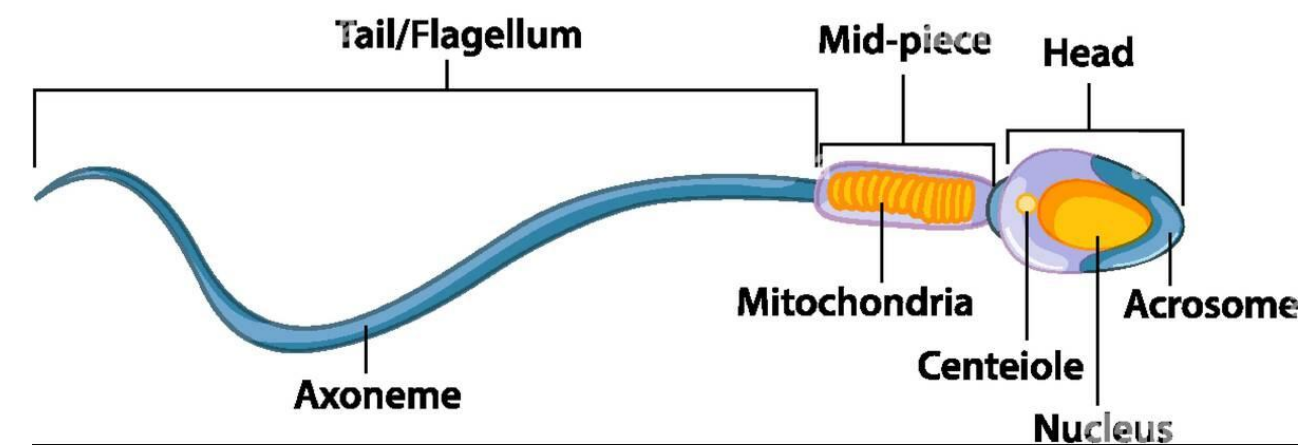
Fertilization Process:

- Sperm meets ovum in female reproductive tract
- Fusion forms zygote
- Zygote develops into embryo

HUMAN EGG (OVUM)



HUMAN SPERM (SPERMATOOZA)



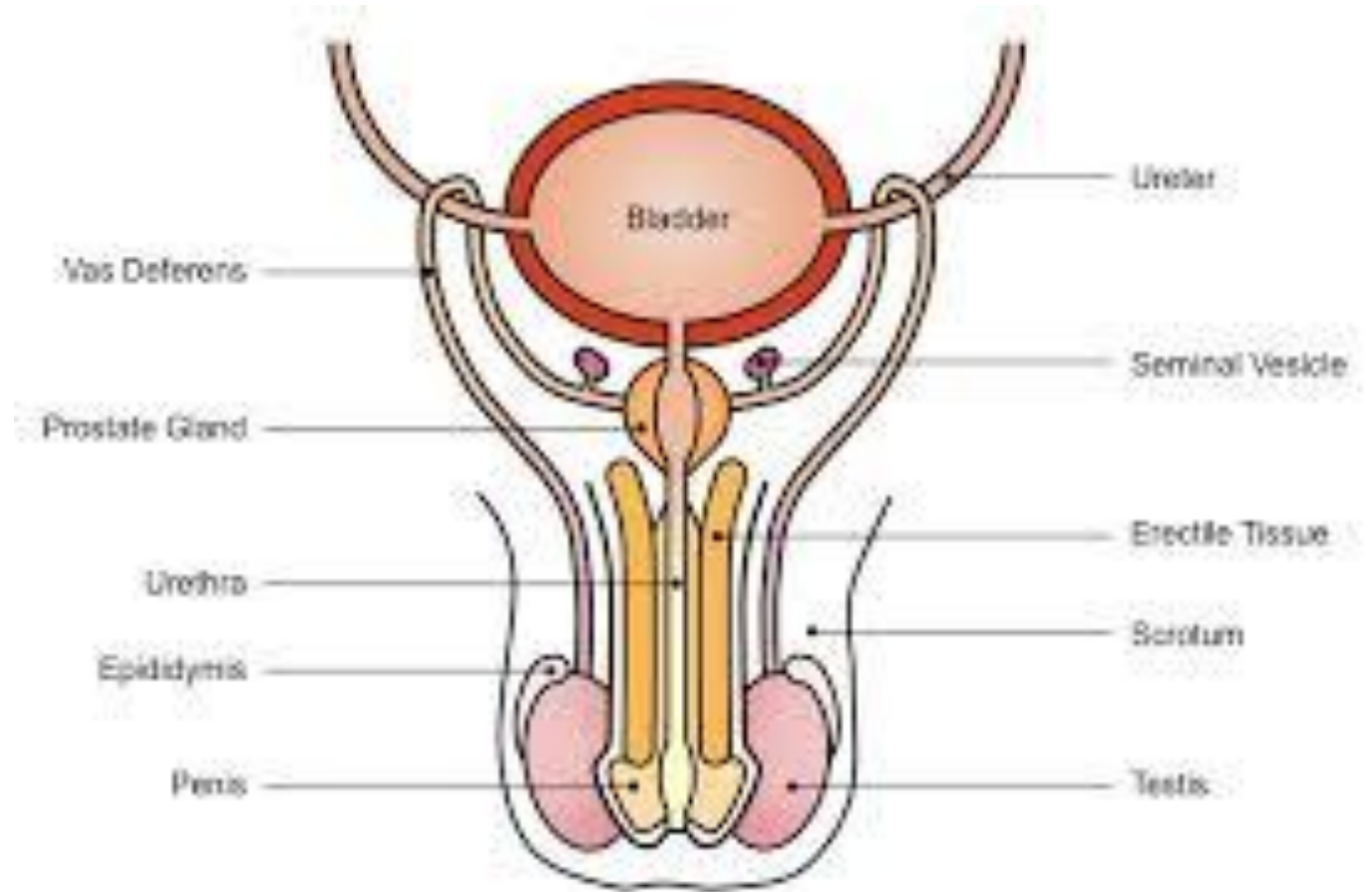
Human Reproductive System - Male

Organs of Male Reproductive System:

- **Testes** - Primary sex organs that produce sperm and testosterone
- **Vas deferens** - Tubes that carry sperm from testes to penis
- **Epididymis** - Storage and maturation site for sperm
- **Penis** - External organ for sperm delivery
- **Prostate gland** - Produces fluid that nourishes sperm

Functions:

- Production of male gametes (sperm)
- Production of male hormone (testosterone)
- Delivery of sperm to female reproductive tract



Human Reproductive System - Female

Organs of Female Reproductive System:

- **Ovaries** - Primary sex organs that produce eggs and hormones
- **Fallopian tubes** - Site where fertilization occurs
- **Uterus** - Organ where embryo develops
- **Vagina** - Birth canal and passage for sperm

Functions:

- Production of female gametes (eggs)
- Production of female hormones (estrogen, progesterone)
- Nurturing and development of embryo/fetus

Female Reproductive System

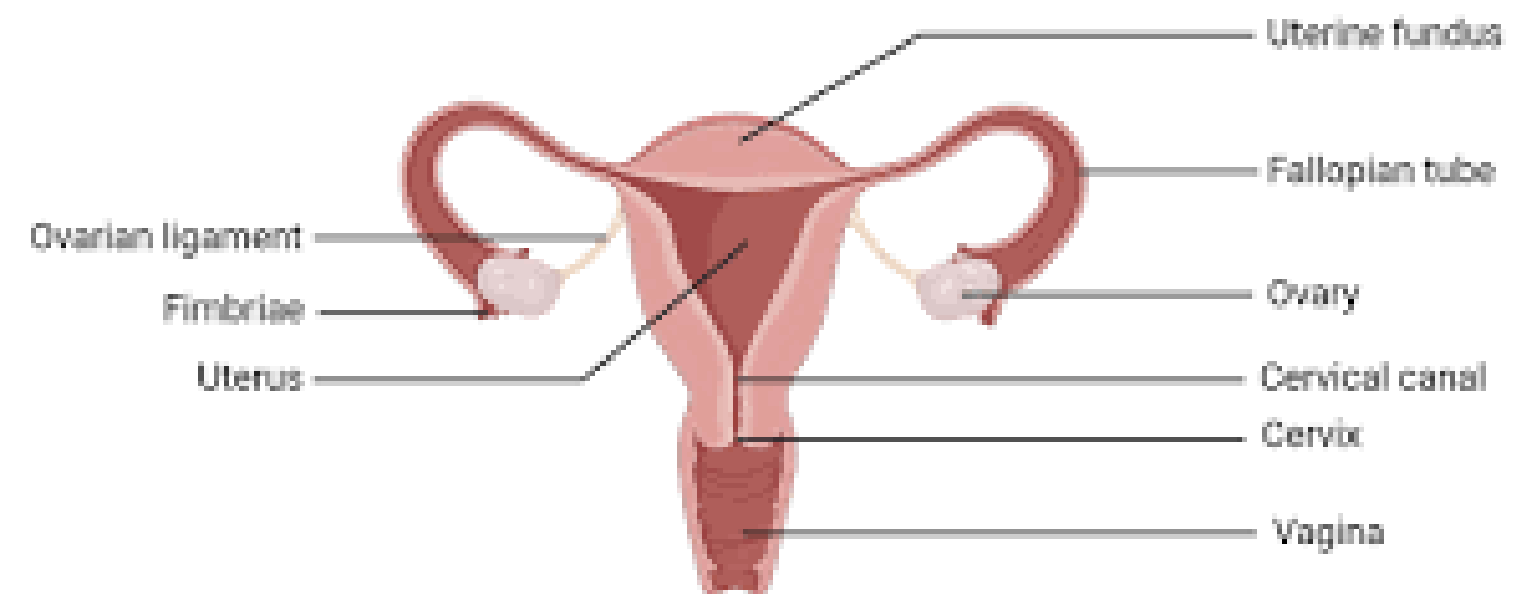


Figure: Structure of Female Reproductive System Designed By: Sagor Aryal. Created with blender.com

Fertilization Process

Types of Fertilization:

1.Internal Fertilization - Occurs inside the female body

- 1.Examples: Humans, birds, mammals
- 2.Higher chances of survival

2.External Fertilization - Occurs outside the body

- 1.Examples: Fish, frogs
- 2.Large numbers of gametes released

Fertilization in Humans:

- Sperm travels through female reproductive tract
- Fertilization occurs in fallopian tubes
- Zygote formed travels to uterus for implantation

Development of Embryo

After Fertilization:

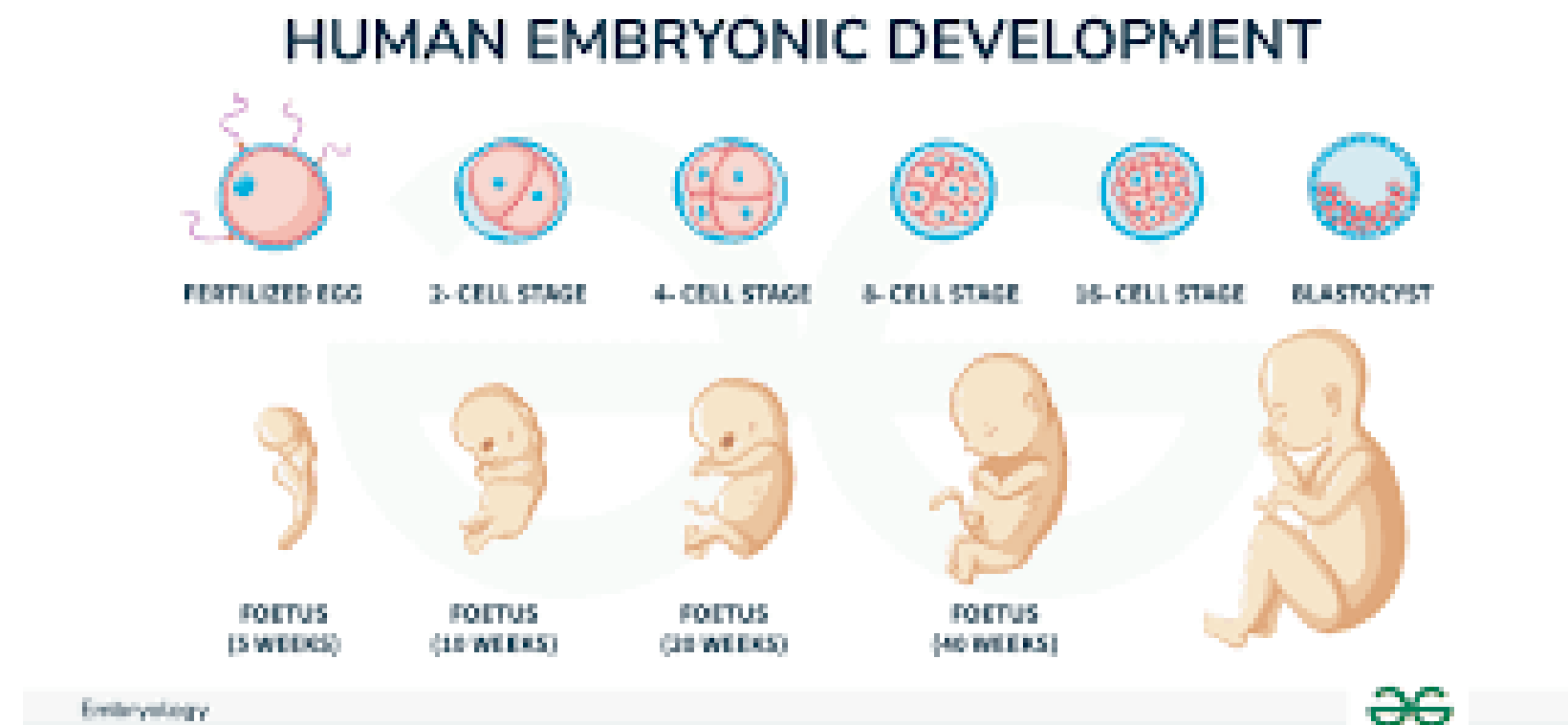
- **Zygote** - Single cell formed after fertilization
- **Embryo** - Early stage of development (first 8 weeks)
- **Fetus** - Later stage of development (after 8 weeks)

Key Processes:

- **Cell division** - Zygote divides to form more cells
- **Differentiation** - Cells become specialized
- **Implantation** - Embryo attaches to uterine wall

Development Timeline:

- Day 1: Fertilization
- Days 2-7: Cell division and travel to uterus
- Week 2: Implantation
- Weeks 3-8: Embryonic development
- Weeks 9-40: Fetal development



Puberty and Adolescence

What is Puberty?

- Period when reproductive organs mature and start functioning
- Usually occurs between ages 11-19
- Controlled by hormones

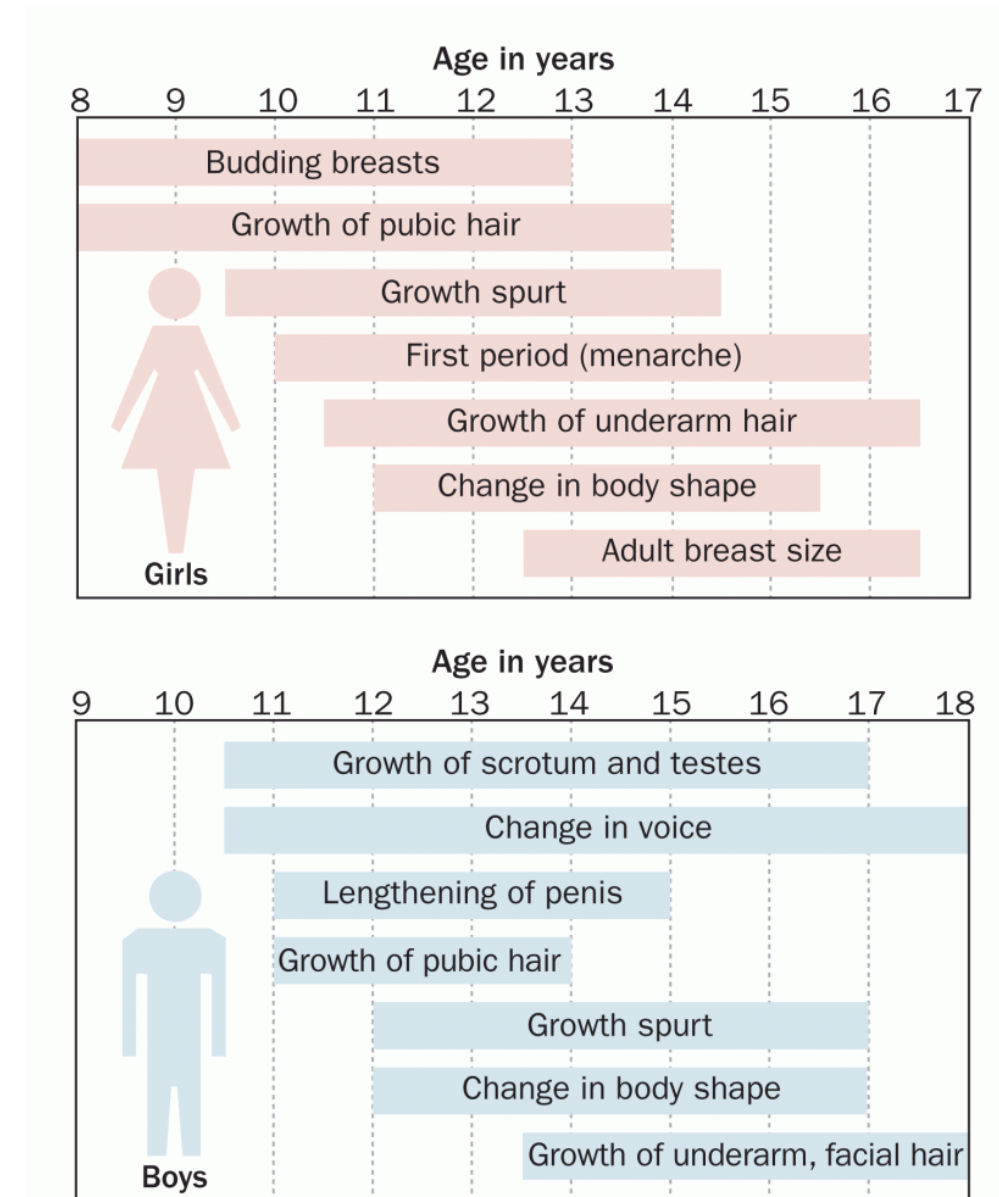
Changes During Puberty:

In Boys:

- Voice becomes deeper
- Facial and body hair growth
- Muscle development
- Height and weight increase
- Reproductive organs mature

In Girls:

- Breast development
- Menstruation begins
- Hip broadening
- Height and weight increase
- Reproductive organs mature



Secondary Sexual Characters

Definition:

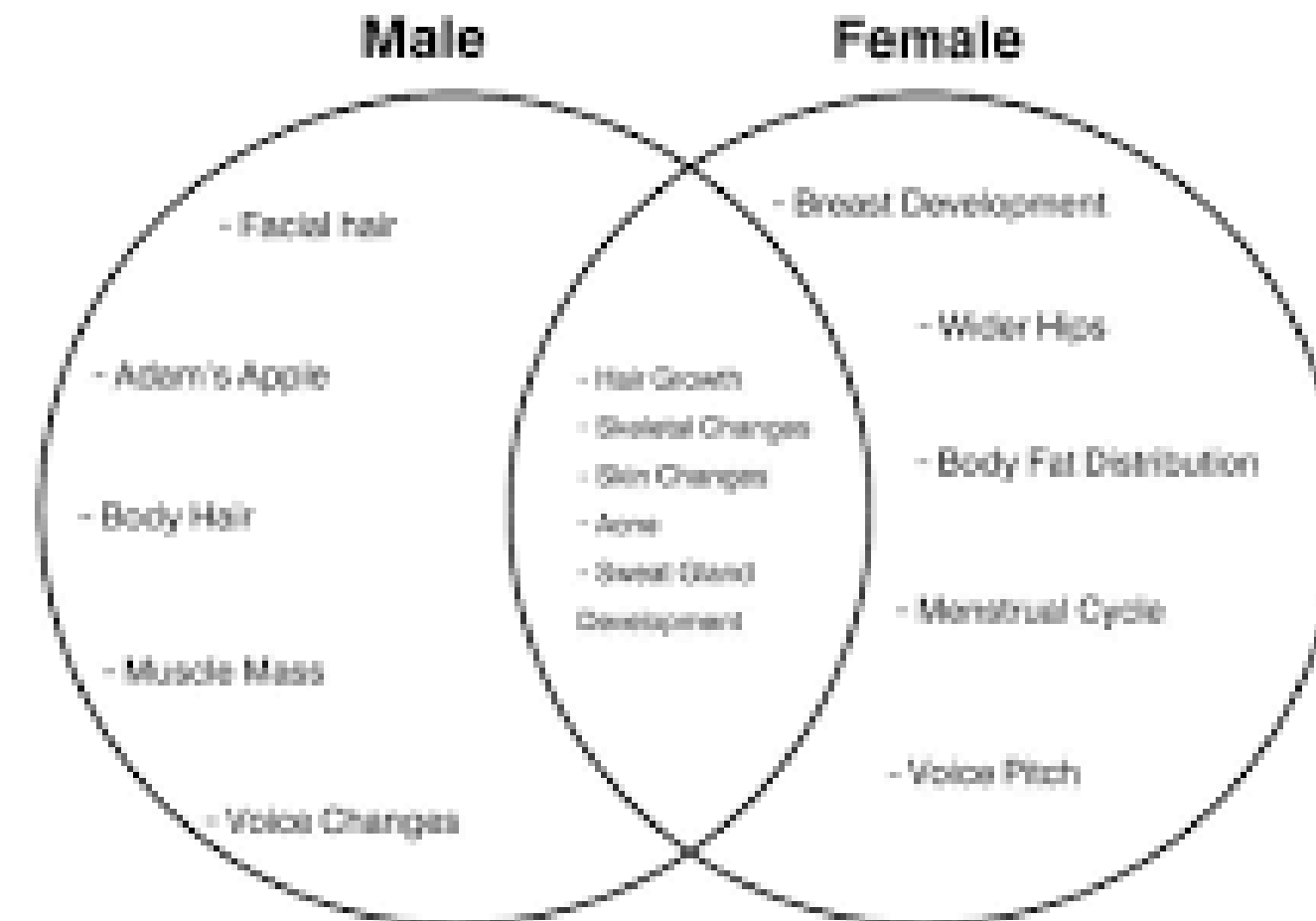
- Features that distinguish males and females but are not directly involved in reproduction
- Develop during puberty due to hormonal changes

In Males:

- Deeper voice
- Facial hair (beard, mustache)
- Broader chest and shoulders
- More muscle mass
- Body hair growth

In Females:

- Higher voice
- Breast development
- Broader hips
- Less body hair
- Fat distribution in hips and thighs



HORMONES

Important Hormones:

Male Hormones:

- **Testosterone** - Controls development of male characteristics
- Produced by testes

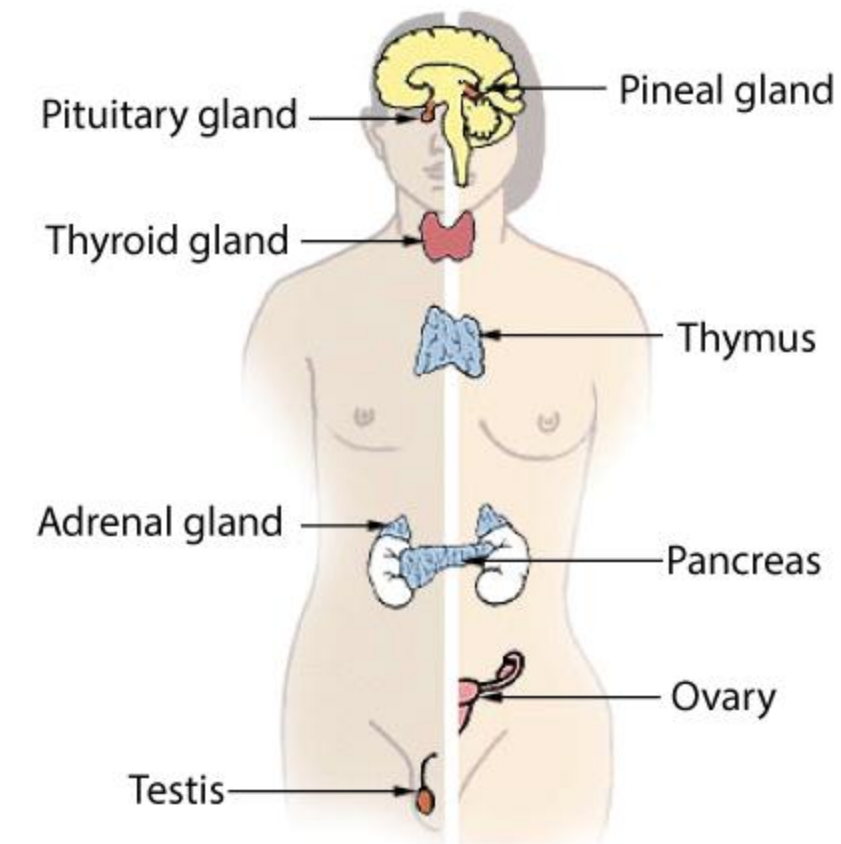
Female Hormones:

- **Estrogen** - Controls development of female characteristics
- **Progesterone** - Maintains pregnancy
- Produced by ovaries

Functions of Hormones:

- Control puberty changes
- Regulate reproductive cycles
- Maintain reproductive health
- Control secondary sexual characteristics

THE ENDOCRINE SYSTEM



ENDOCRINE SYSTEM

Brain Glands

- **Pineal** – Controls sleep.
- **Pituitary** – Controls growth and other glands.

Neck

- **Thyroid** – Controls body energy and growth.

Chest

- **Thymus** – Helps in building immunity.

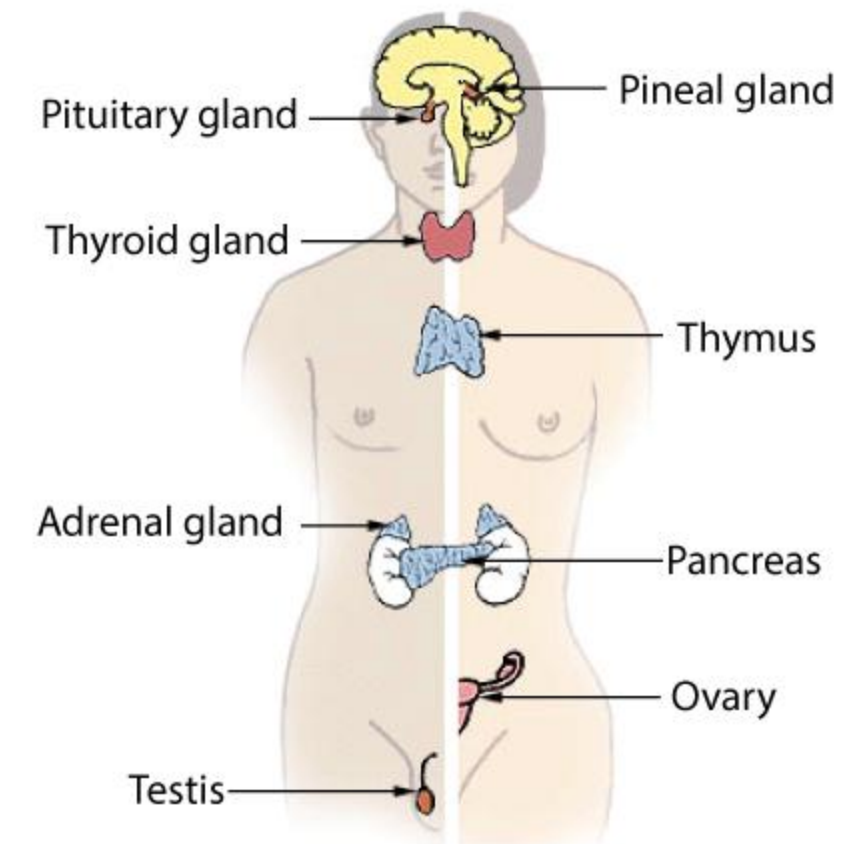
Abdomen

- **Adrenal** – Helps in fight or flight (emergency).
- **Pancreas** – Controls blood sugar.

♂ Reproductive Organs

- **Testis (Boys)** – Makes testosterone (voice, muscles).
- **Ovary (Girls)** – Makes estrogen (breasts, periods).

THE ENDOCRINE SYSTEM



REPRODUCTIVE HEALTH

Importance of Reproductive Health:

- Proper hygiene and care
- Regular health check-ups
- Balanced nutrition
- Exercise and healthy lifestyle

Common Issues:

- Infections
- Hormonal imbalances
- Infertility problems

Prevention:

- Maintain personal hygiene
- Avoid unsafe practices
- Seek medical advice when needed
- Follow healthy lifestyle

